MSE DRONE LEG QUALIFICATION

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PROBLEM STATEMENT

When a Part Needs a Second Life...

How do engineers reconstruct, optimize, and qualify an existing component when the original design is unavailable or no longer meets its intended purpose?

Our Objective

Experience the engineering design process by reconstructing, optimizing, manufacturing, and validating an existing mechanical component while preserving its intended geometry, functionality, and manufacturability.

Case Study

A landing foot for a 2kg UAV was selected as the platform to apply this workflow.

ENGINEERING DESIGN PROCESS

Existing Part

Reverse Engineer

Digital Reconstruction

Optimization

Manufacturing

Testing & Qualification

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CASE STUDY

Concept

Apply the complete engineering design process to a challenge problem involving the reverse engineering & qualification of a UAV landing leg that requires reconstruction, optimization, and manufacturing.

Important Factors

- Support takeoff and landing in varied landing conditions (*e.g., grass, Soil, Concrete*)
- Ensure onboard packages remain safe on a heavy impact
- Economical production while maintaining structural integrity
- Achieve through metrology, design, manufacturing, and testing

METROLOGY

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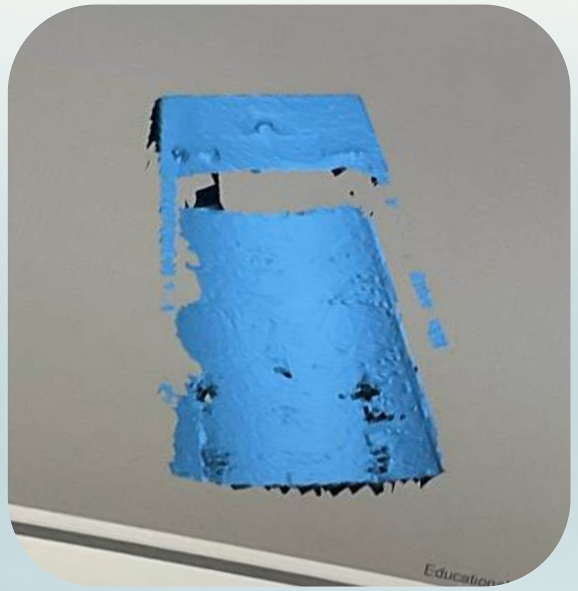
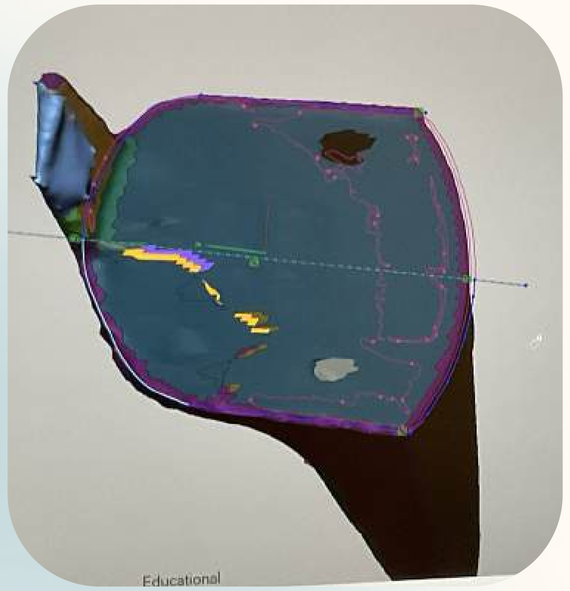
Object Of Interest



3D Scan



Metrology Extraction

CAD Reconstruction

REVERSE ENGINEERING

The process of replicating a design by physically measuring an existing item to develop the technical data necessary to reproduce the item functionally and dimensionally.

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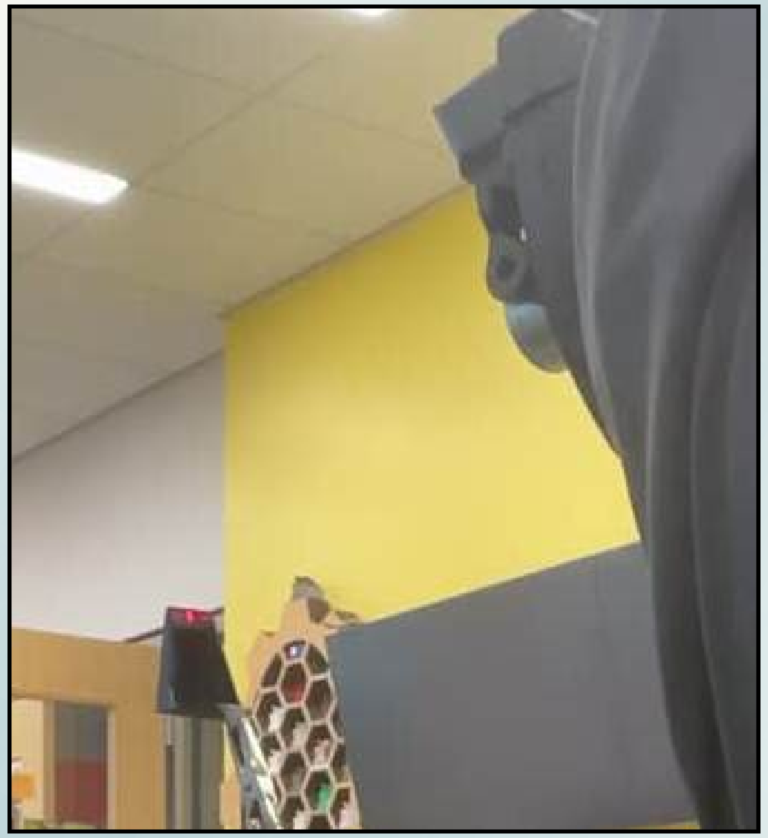
3D SCANNING

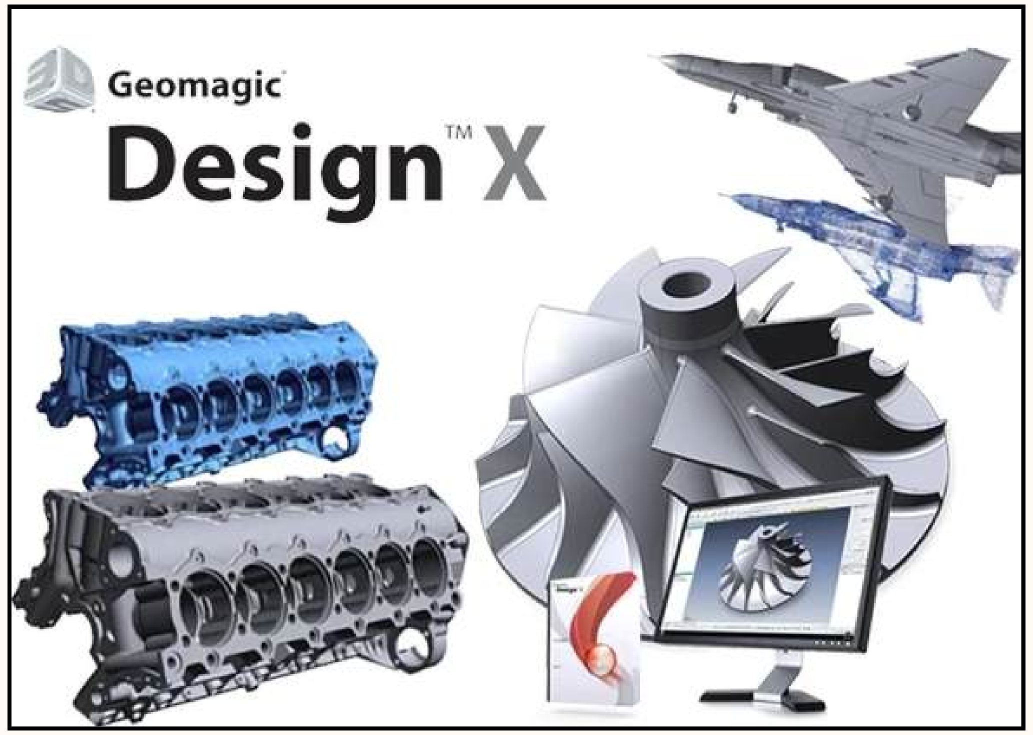
Tools & Processes

Hexagon Romer Absolute Arm

Used for taking 3D scans of various objects

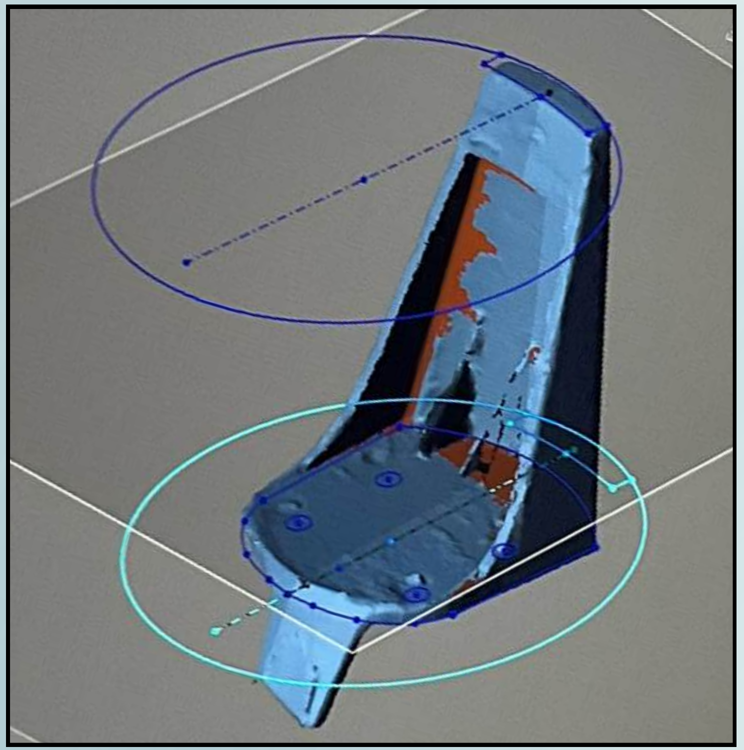


The Tedious 3D Scanning Process



Geomagic Design X

Used for compiling, editing, and extracting object features



The Lengthy 3D Mesh → CAD Process

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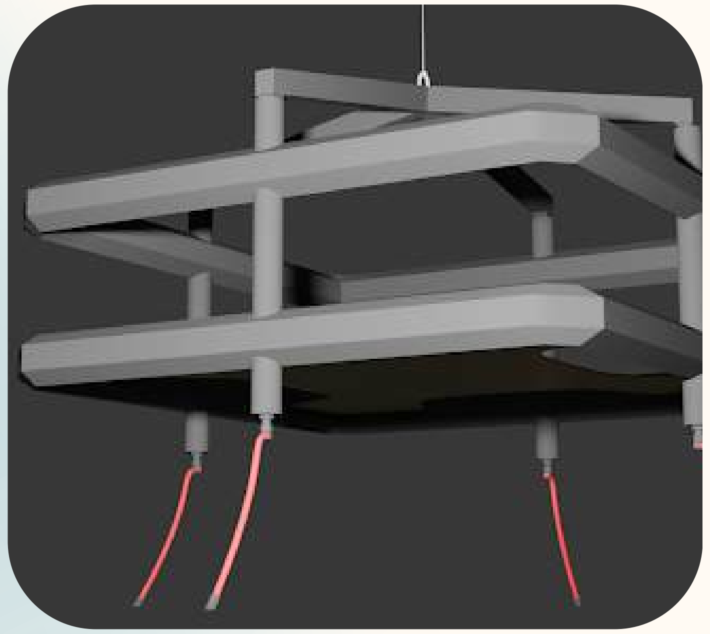
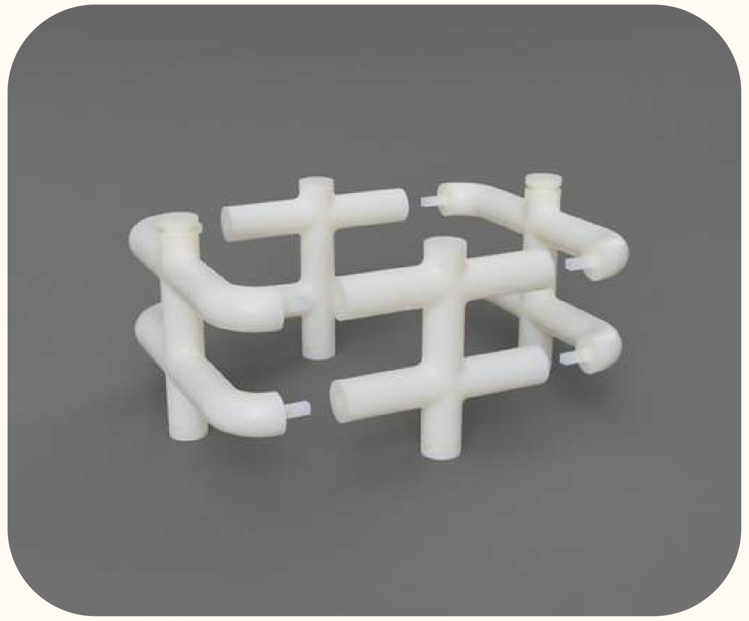
DESIGN

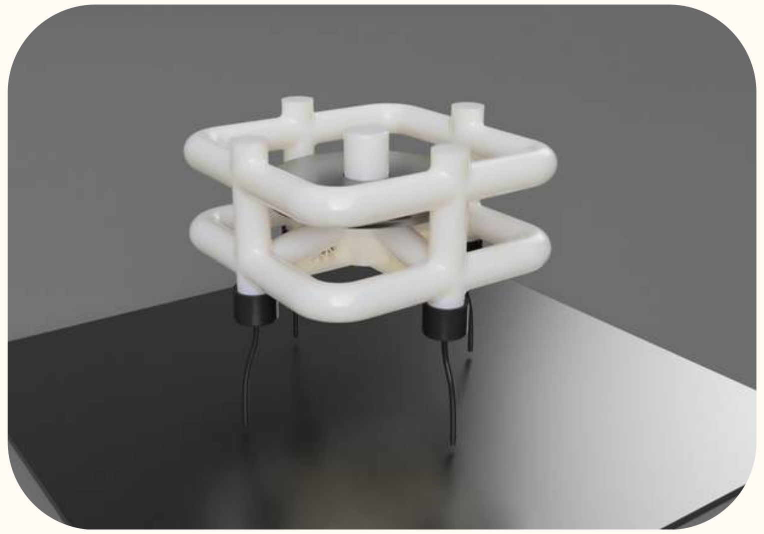
CHASSIS DESIGN

Create a method to support a realistic testing mode

- Support the intended 2 kg total mass
- Allow for quick customization
- Maintain structural integrity with less material
- Maintain protective capabilities in the event of damage
- Inspired by dimensions of Industrial drones







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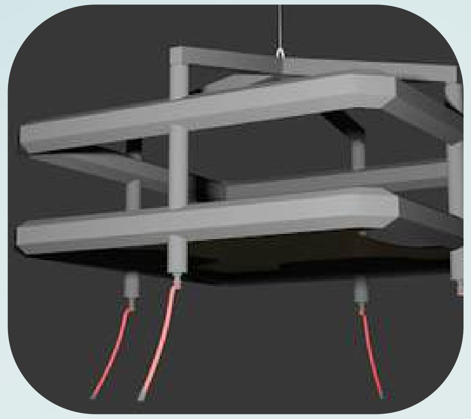
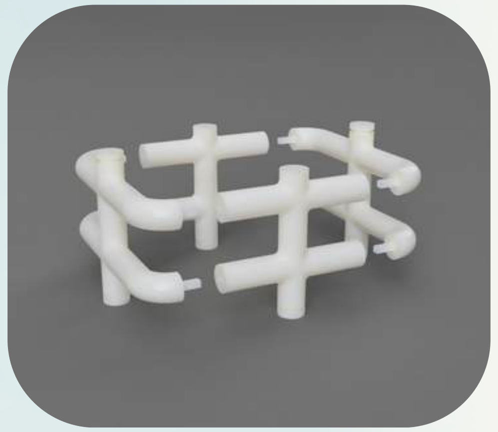
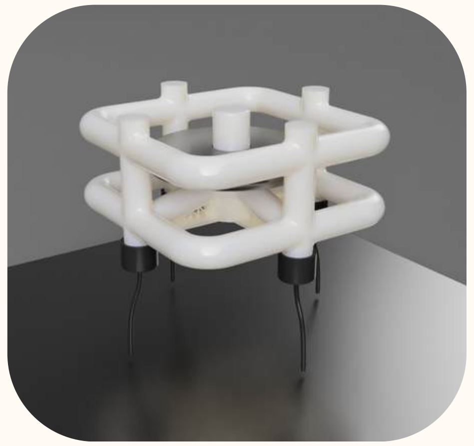
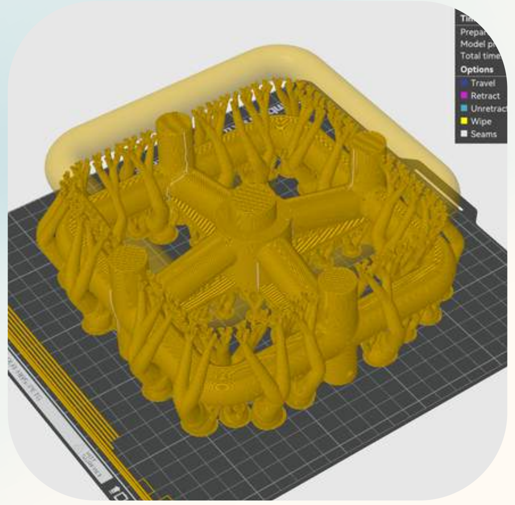
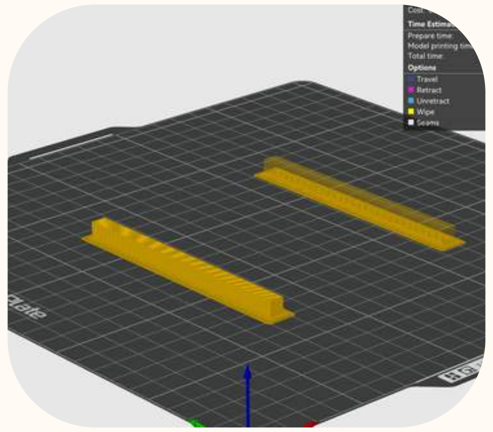
ITERATIVE DESIGN

Build better and timely with an informative fail-fast approach




FINAL DESIGN

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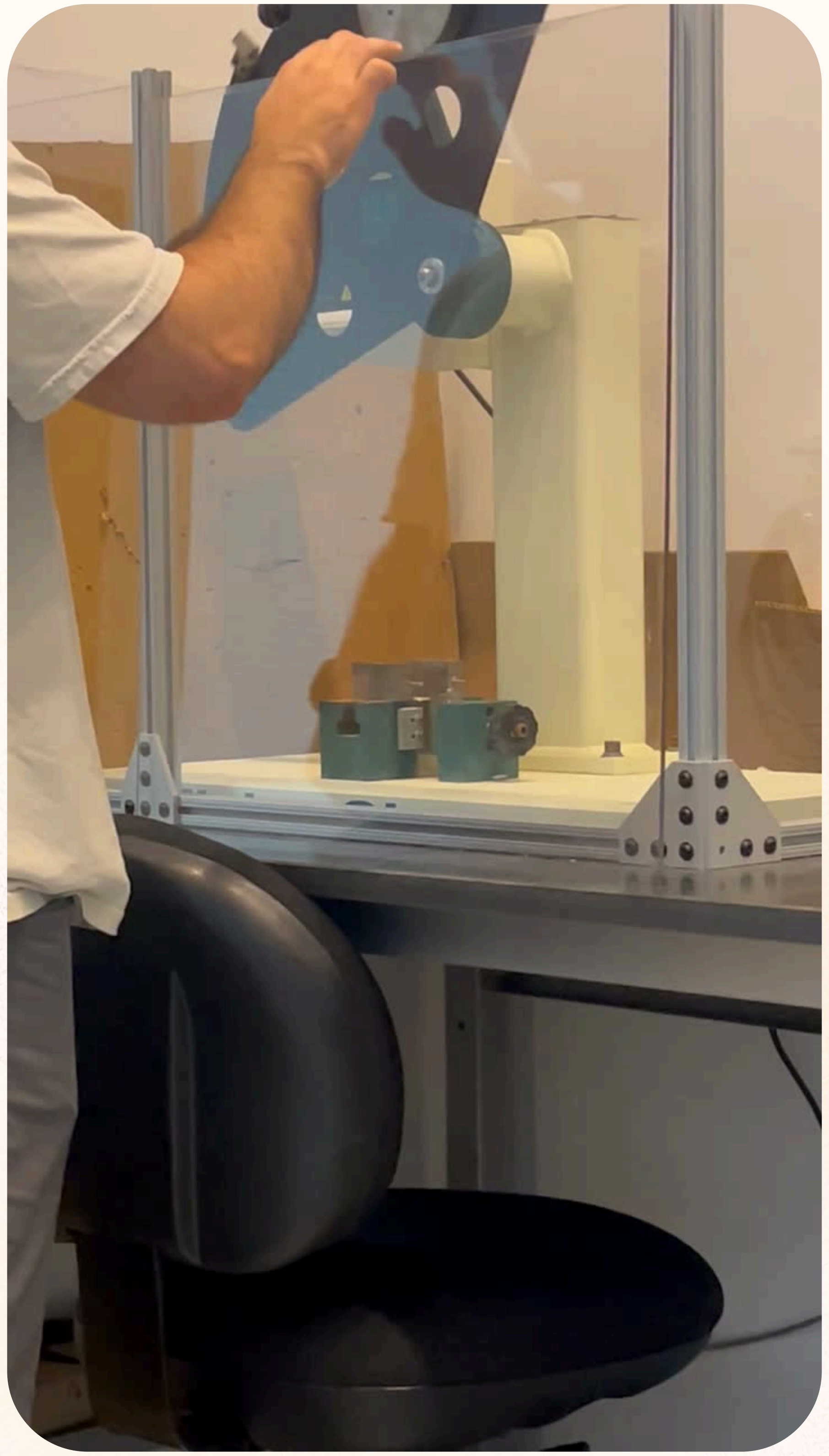
DESIGN EXPERIMENTATION

WHAT IS A CHARPY TEST?

This is an impact test used in materials science, used to determine the amount of energy absorbedin a material (measured in joules) when it is impacted upon or fractured. This helps measure the material's toughness.

This is a clip of one of the tests, we worked with a PHD student of Dr. Valentino's named Eliott, who is trained in the Dieter Lab to conduct the tests for us, while we recorded the data.



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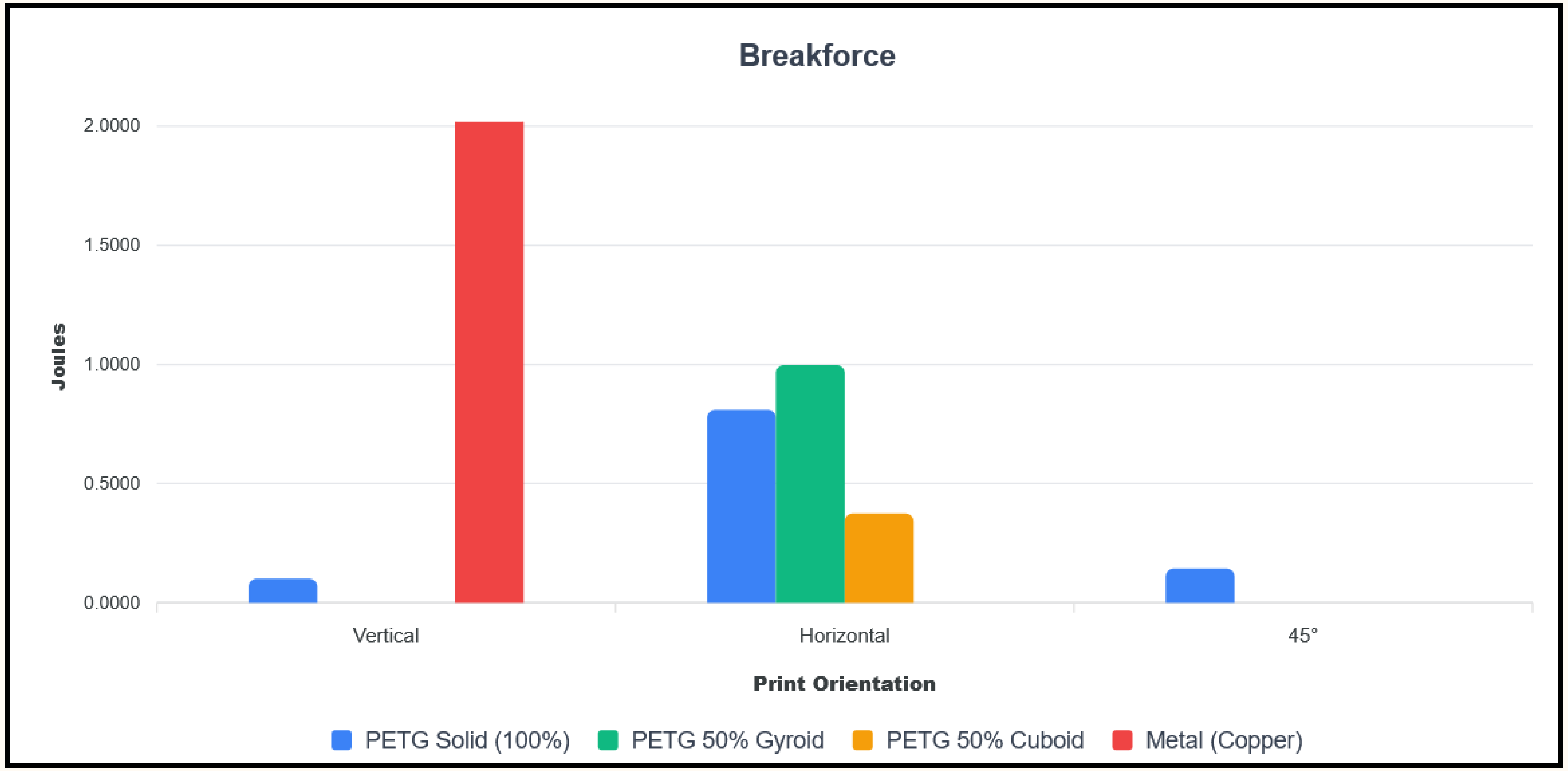
CHARPY IMPACT TEST RESULTS



CALIBRATION	MATERIAL	ORIENTATION	BREAKFORCE	INFILL
-0.0705 JOULES	PETG	VERTICAL	0.1010 JOULES	SOLID
	PETG	HORIZONTAL	0.8094 JOULES	SOLID
	PETG	45°	0.1431 JOULES	SOLID
	PETG	HORIZONTAL	0.9976 JOULES	50% GYROID
	PETG	HORIZONTAL	0.3738 JOULES	50% CUBOID
0.2745 JOULES	COPPER	N/A	161.36 JOULES	SOLID

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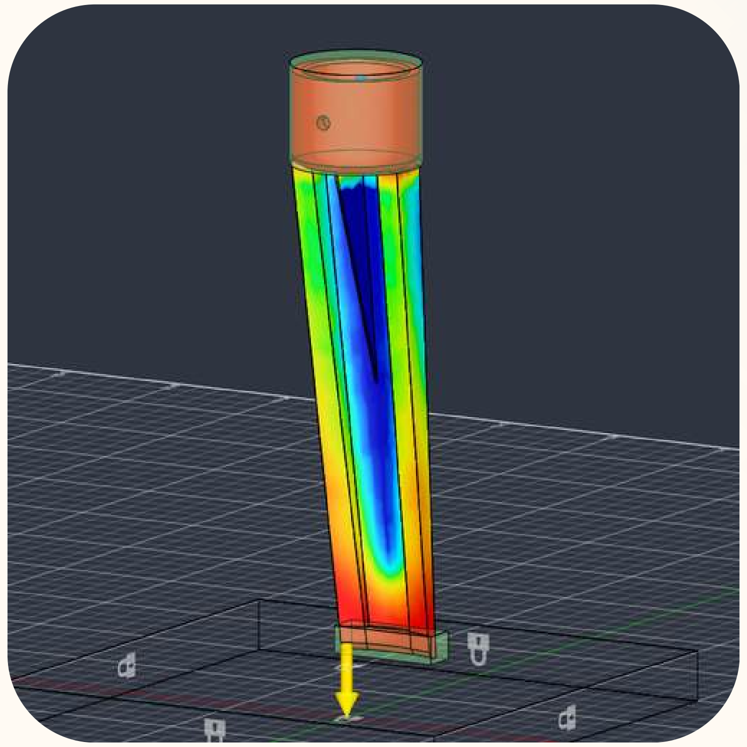
LEG DESIGN

Produce an economical, lightweight, and durable design



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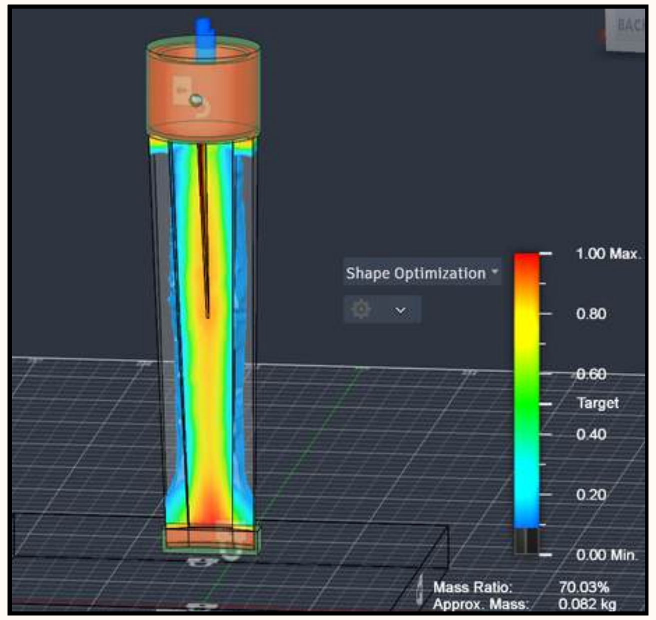
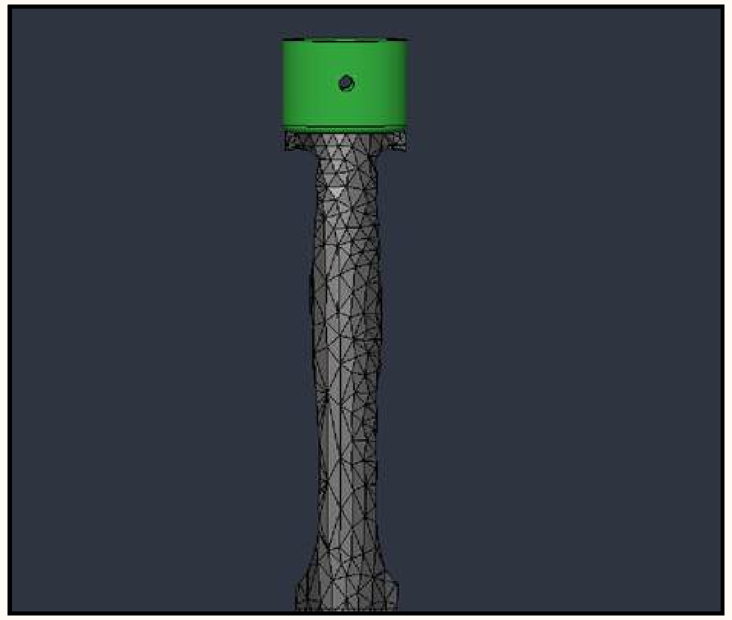
SHAPE OPTIMIZATION

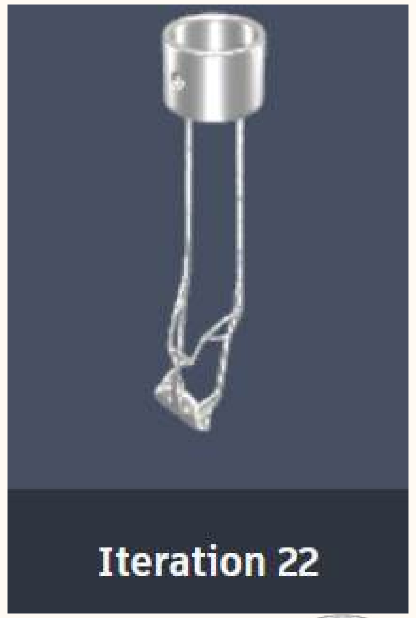
Identifies the most efficient material distribution under the applied loads and constraints





GENERATIVE DESIGN

Generates multiple manufacturable design concepts while preserving critical mounting regions.



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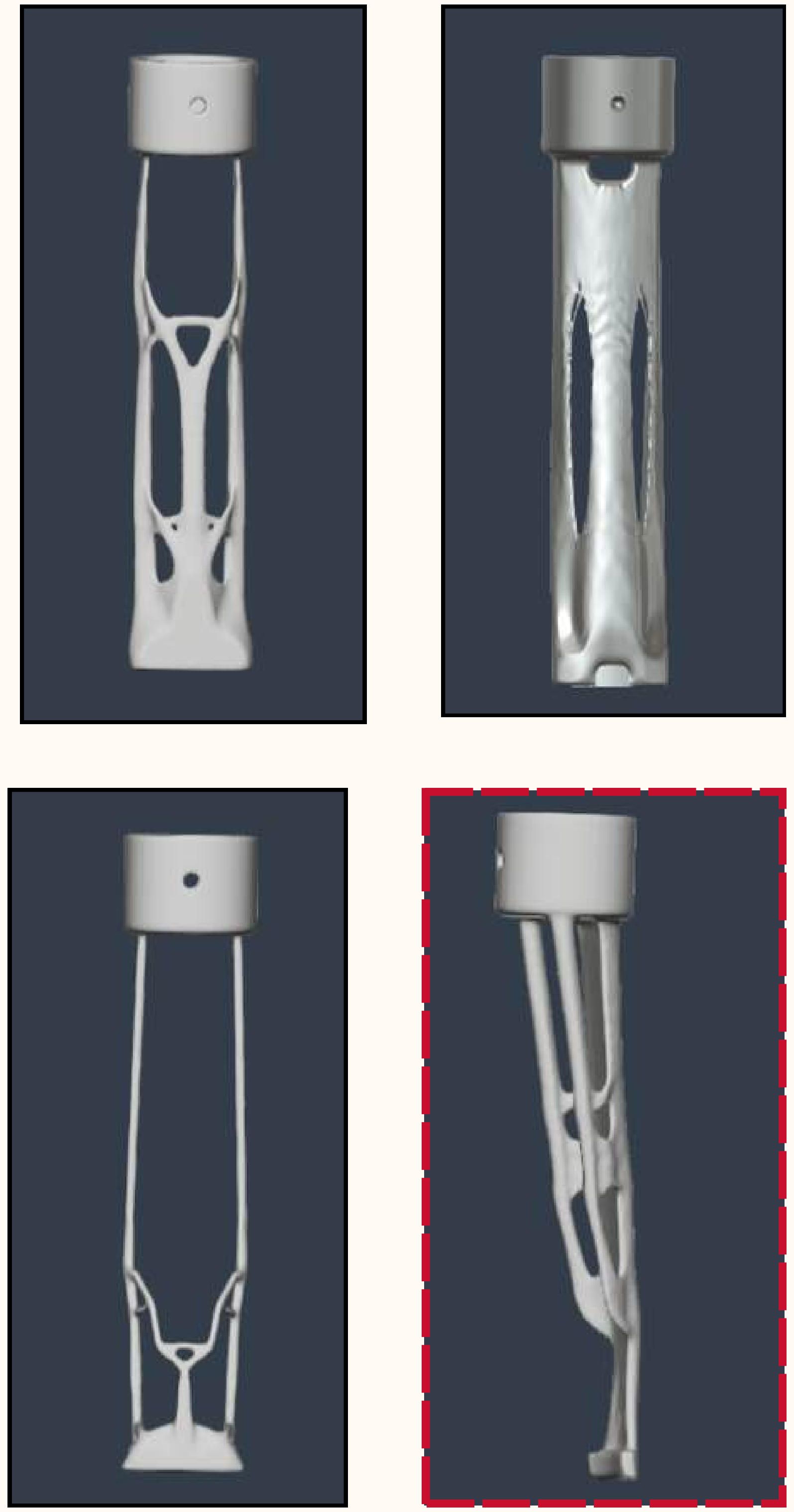
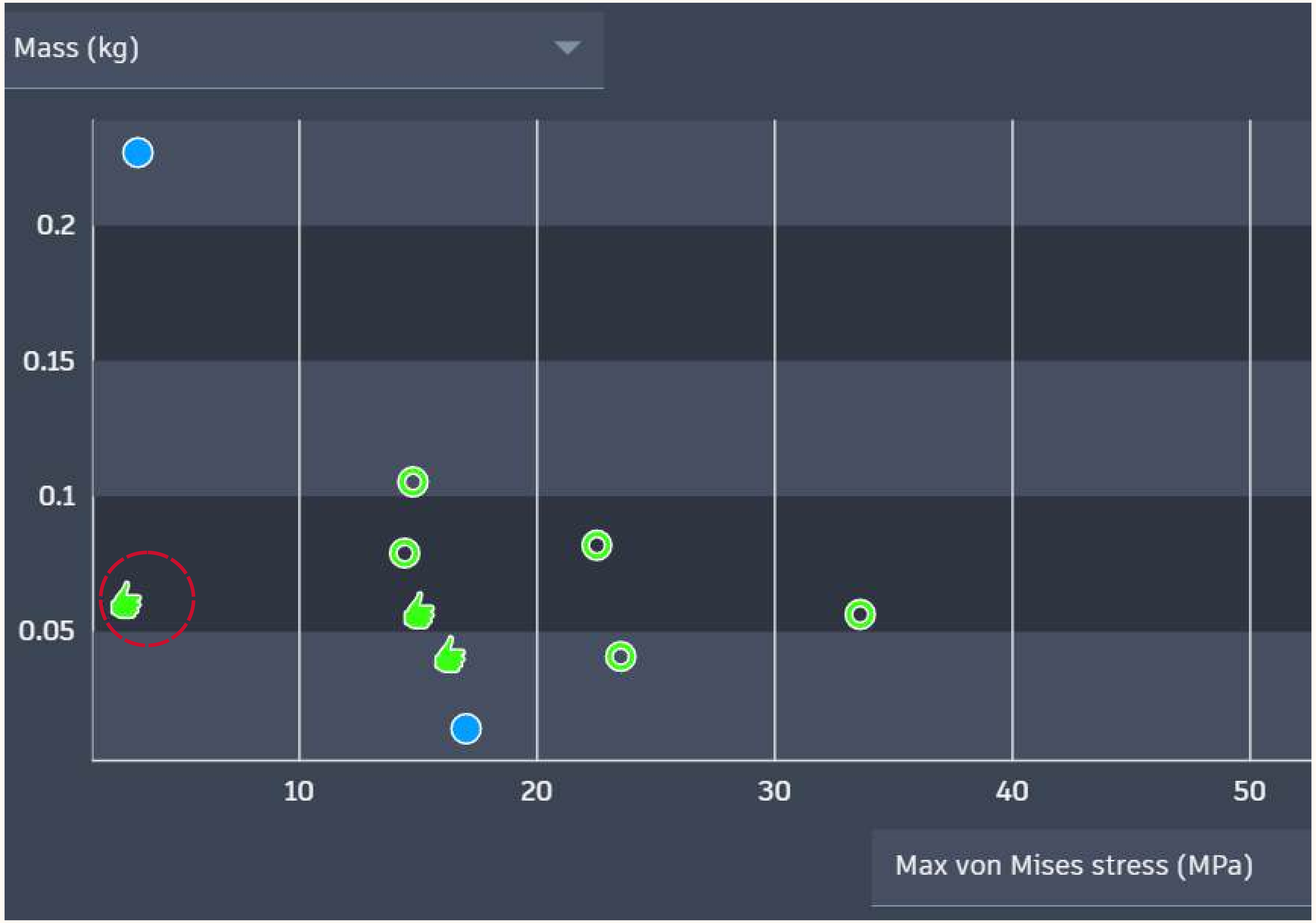
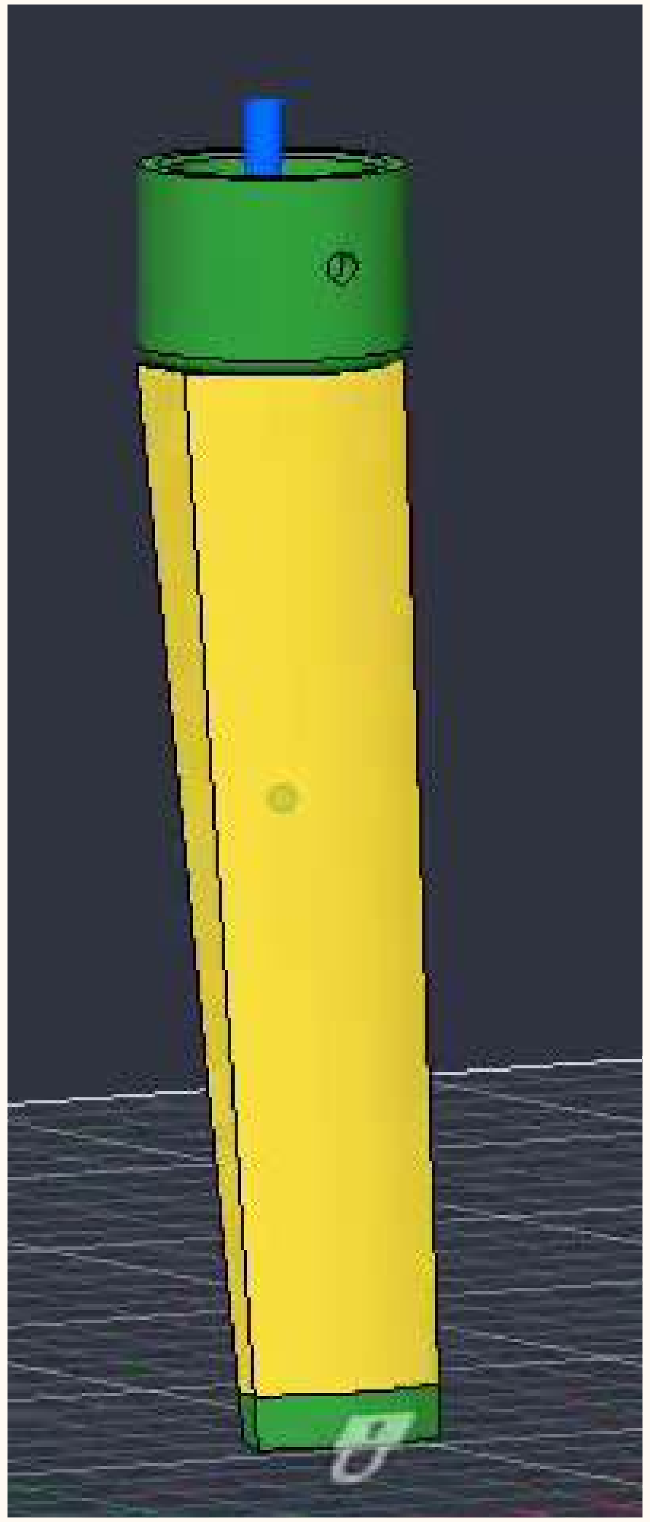
GENERATIVE DESIGN

Generates multiple manufacturable design concepts while preserving critical mounting regions







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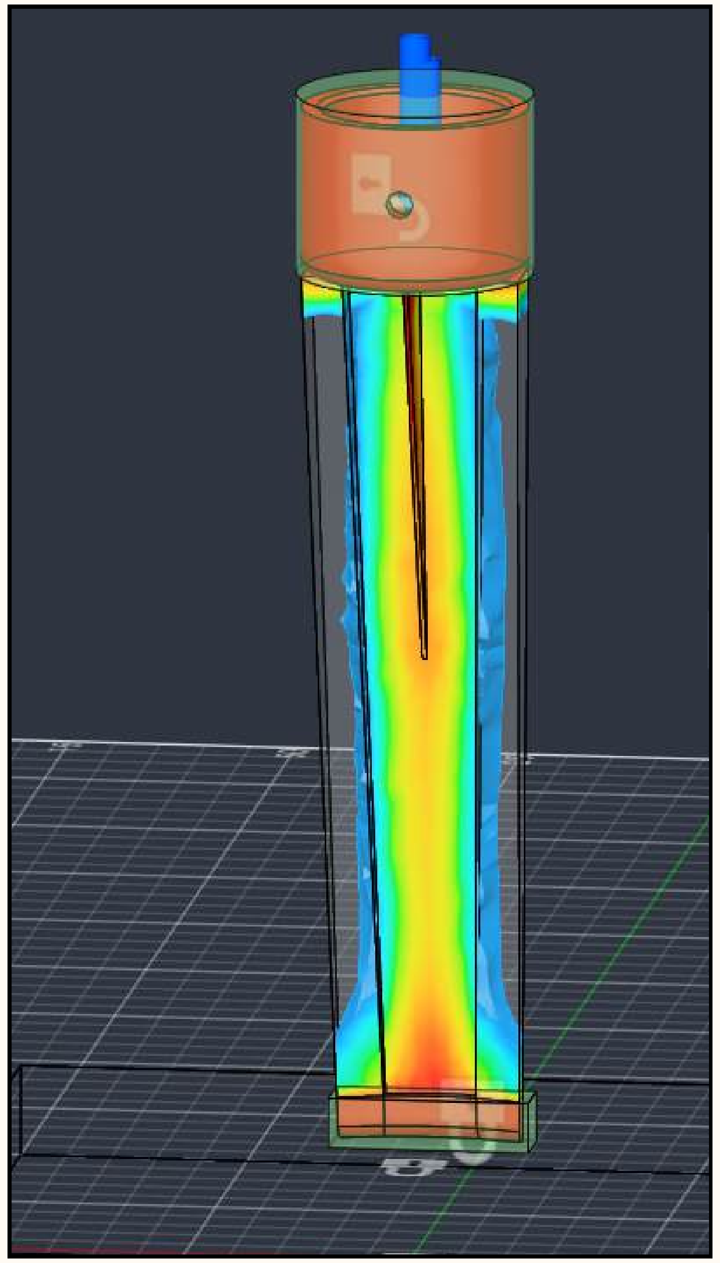
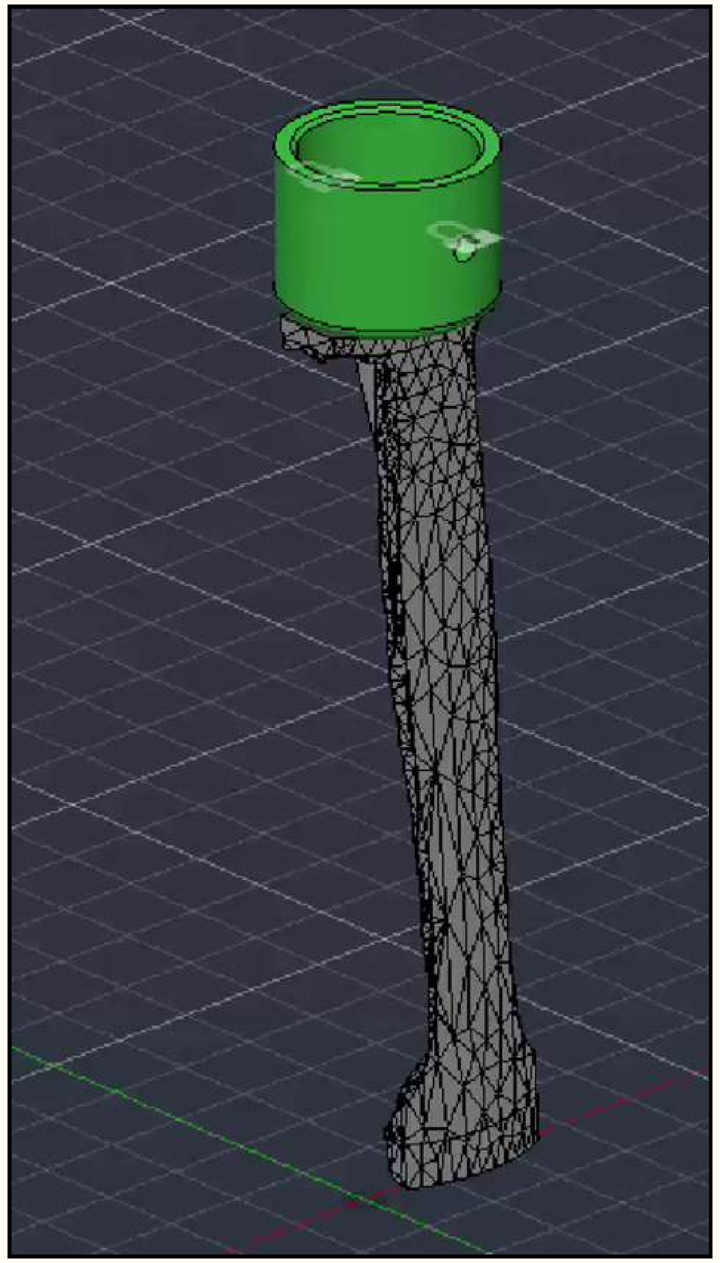
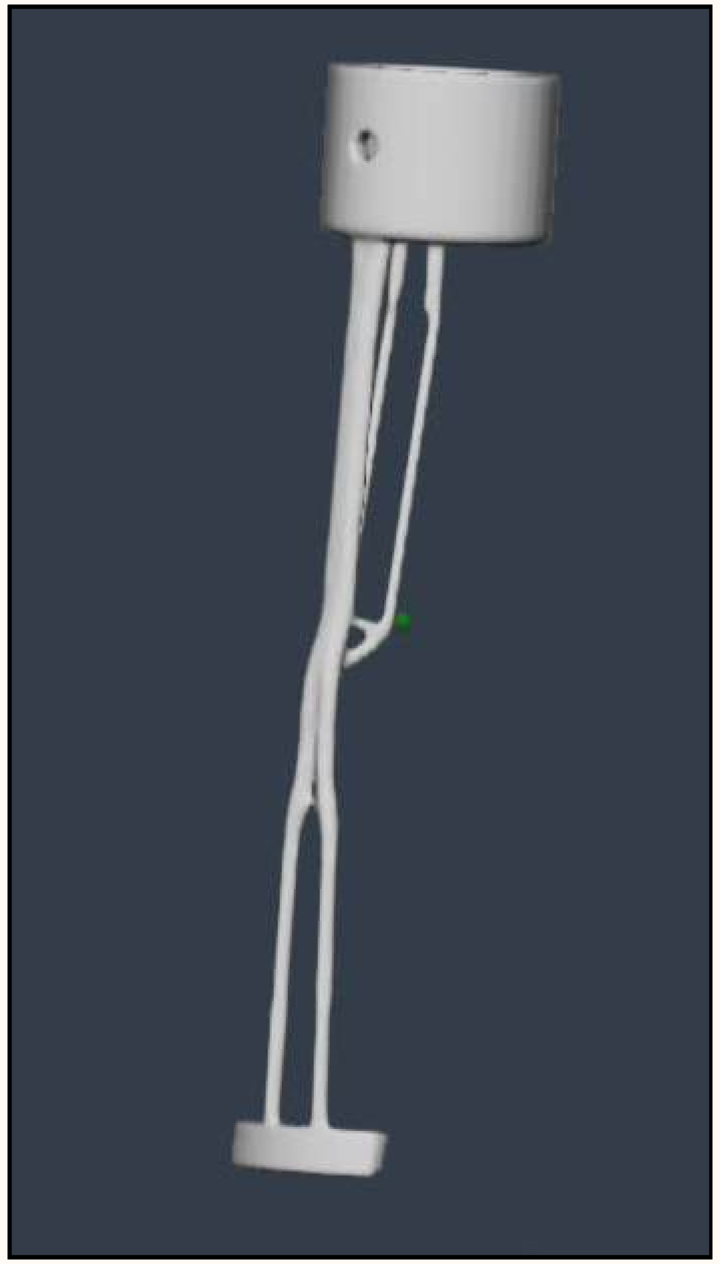
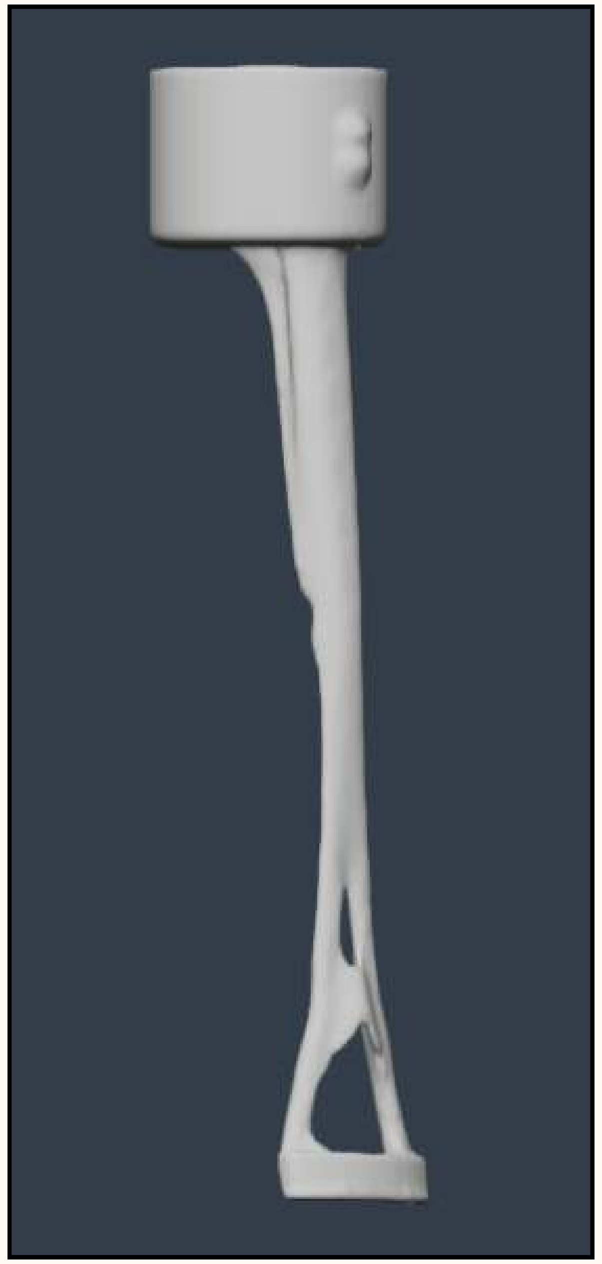
HYBRID OPTIMIZED DESIGN

Combines the structural efficiency of the topology-optimized mesh with the manufacturable geometry of the selected generative design









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COMPUTATIONAL MODELING

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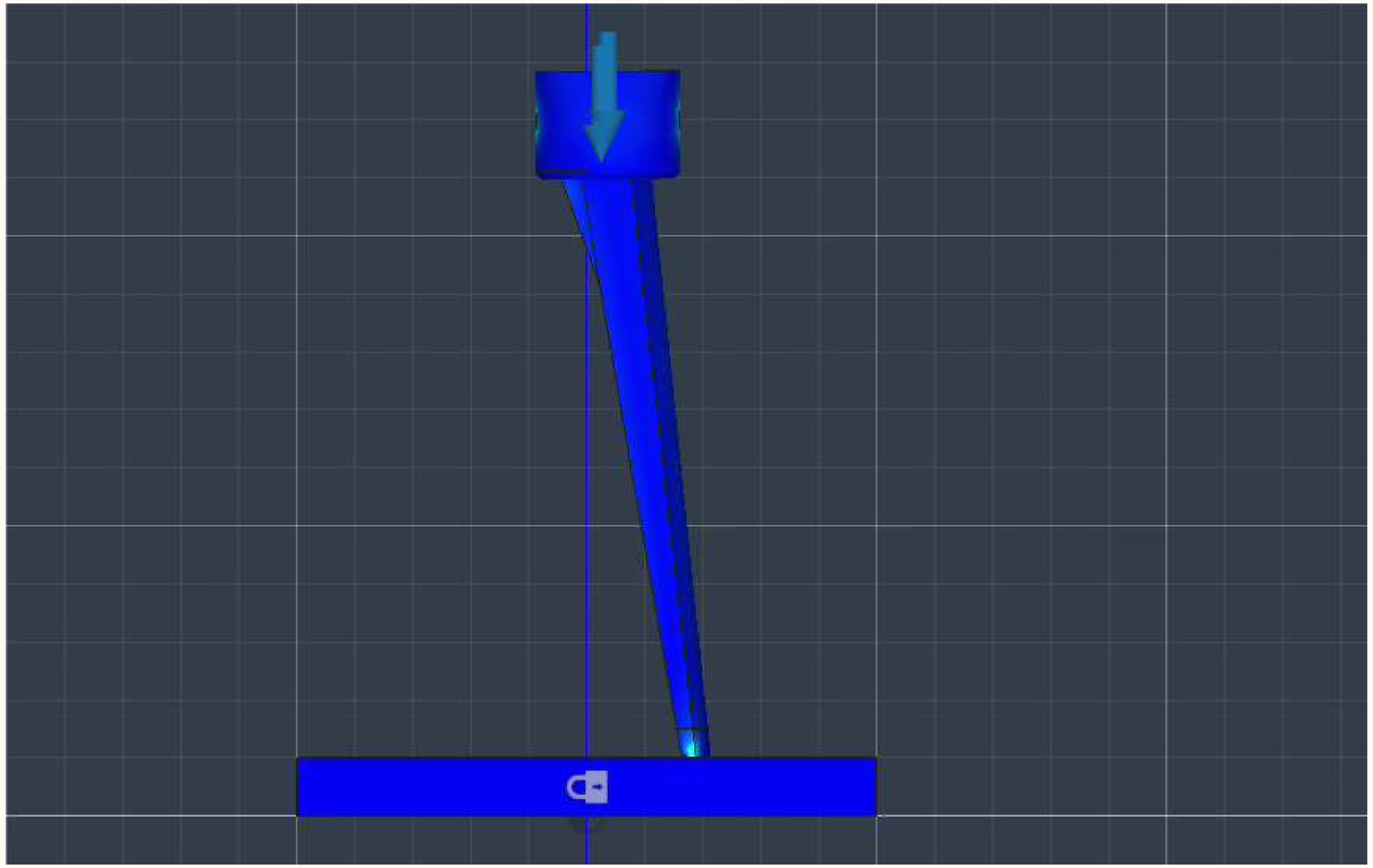
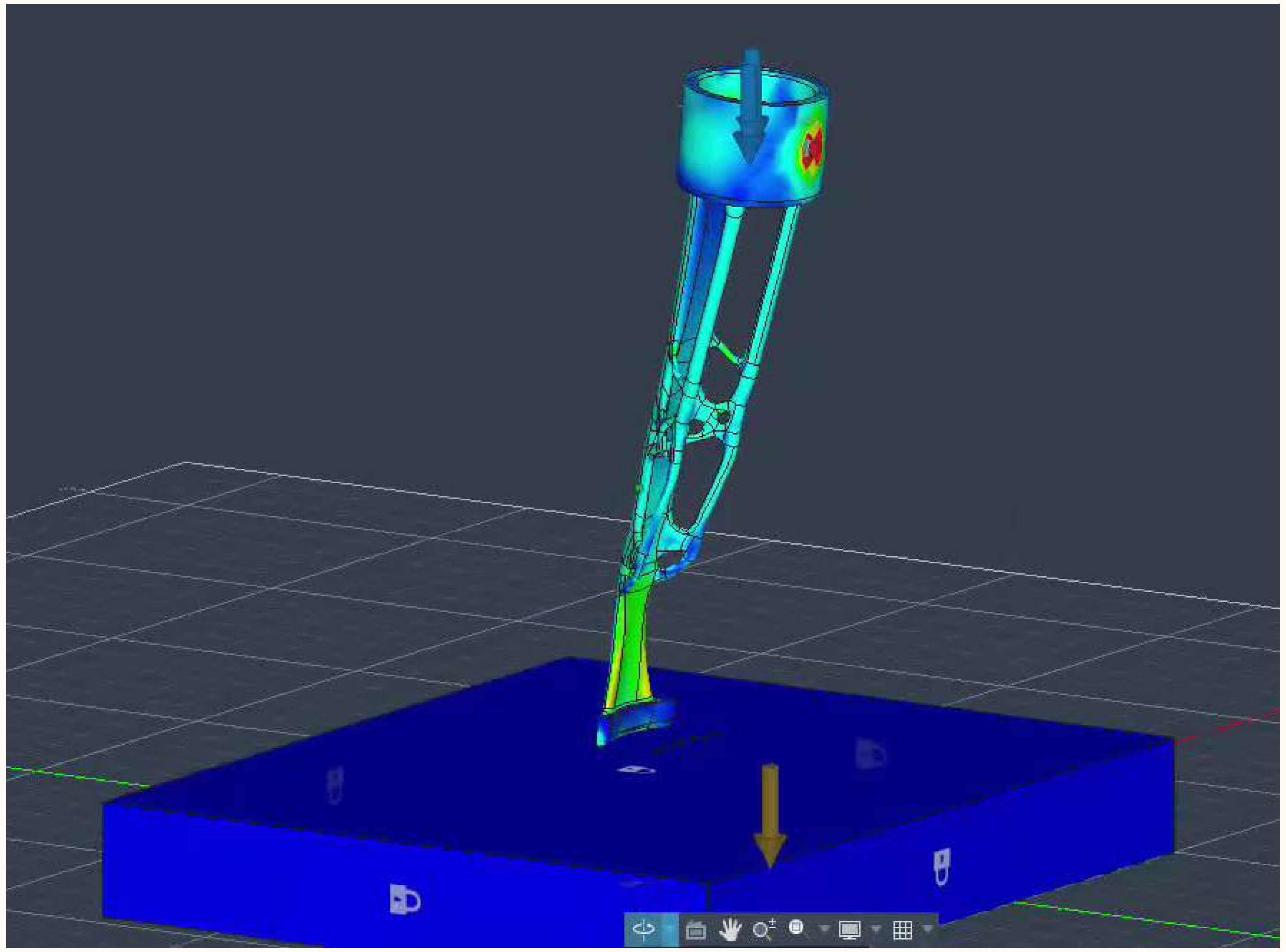
STATIC STRESS MODELING

Model performance under a static loading environment



Original Leg
58 N



Optimized Leg
58 N

DYNAMIC EVENT MODELING

Transitioned from static stress analysis to a dynamic event simulation to capture transient force effects

Dynamic Event	Static Stress
Analyzes short, fast, and violent events like crashes, drops, or hits.	Calculates safe loads, displacement, and stress under steady or slowly applied forces.
Accounts for mass, velocity, acceleration, inertia, and fast-changing contact between parts.	Ignores inertia, momentum, and acceleration because the load changes very little over time.
Requires non-linear stress-strain curve	Does not require stress-strain curve

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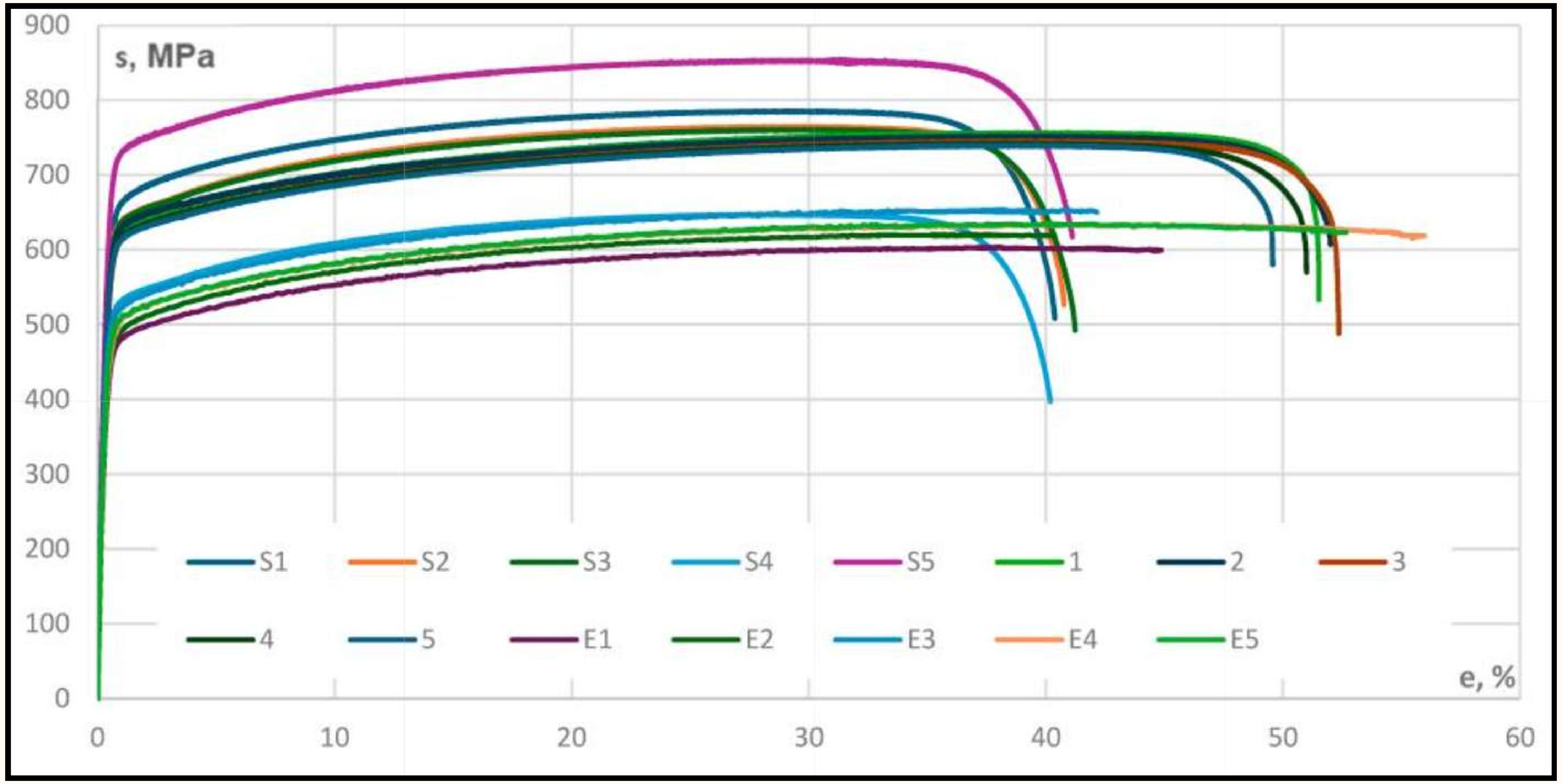
DATA MODELING BLOCKS

Autodesk Fusion’s default library lacks the non-linear stress-strain curve data required for Stainless Steel 316L impact simulations

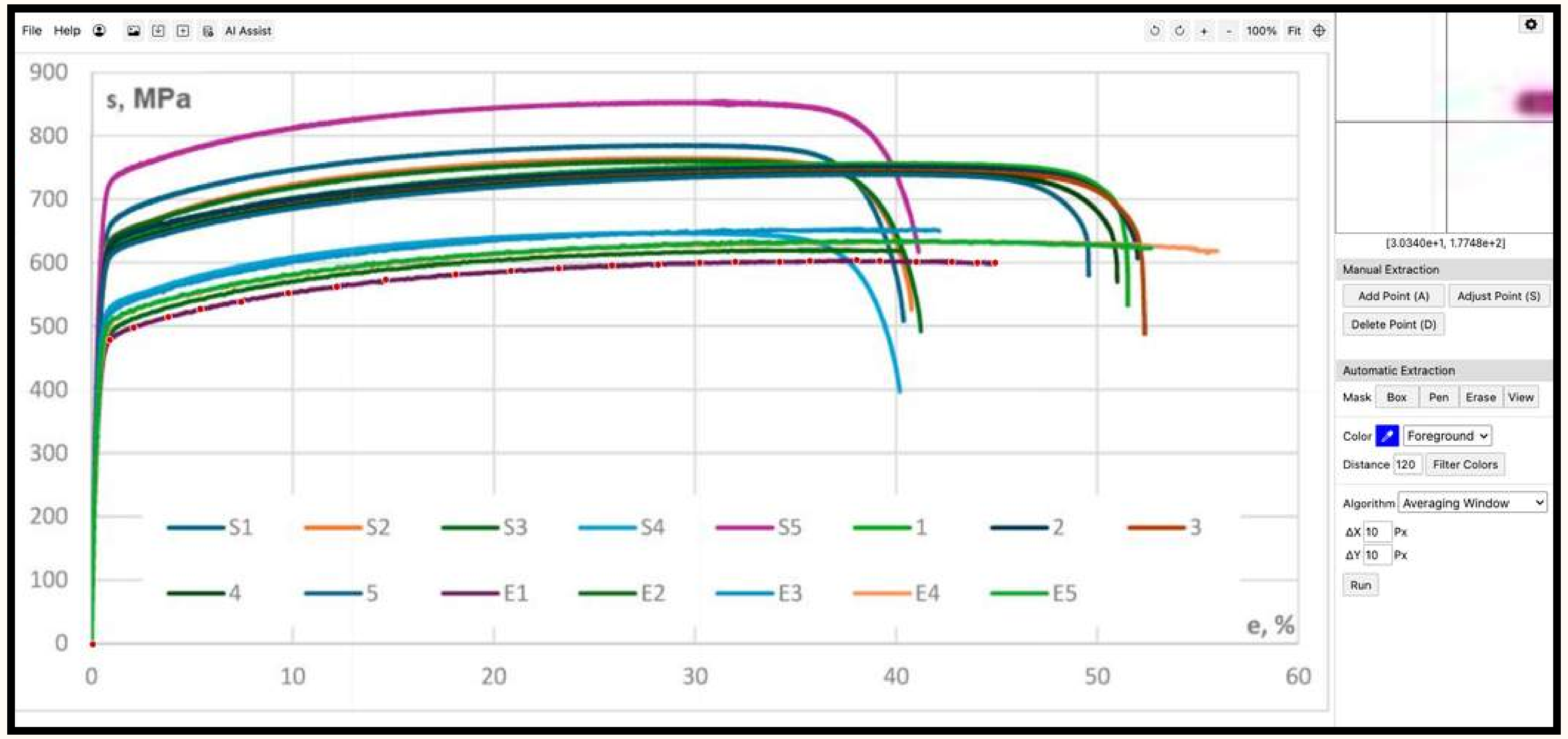
- **Resolution:** Sourced a peer-reviewed research paper containing empirical data detailing the exact stress-strain curves of SLM-printed SS316L.



Sumanariu, Constantin Alex et al. “Comparative Analysis of Mechanical Properties: Conventional vs. Additive Manufacturing for Stainless Steel 316L.” Materials (Basel, Switzerland) vol. 17,19 4808. 29 Sep. 2024, doi:10.3390/ma17194808

DATA EXTRACTION

Digitization



Data Conversion

```
[41]: import pandas as pd
      import numpy as np

      df = pd.read_csv('/Users/justinokoli/Downloads/Nonlinear Data from WebPlotDigitizer.csv', header=None)

      strain_dec = df[0]/100
      true_strain = np.log(1 + strain_dec)
      true_stress = df[1] * (1 + strain_dec)
      df_true = pd.DataFrame({
          'True Strain': true_strain.round(6),
          'True Stress (Mpa)': true_stress.round(6)
      })

      print(df_true)
```

	True Strain	True Stress (Mpa)
0	0.000000	0.000000
1	0.008587	483.481563
2	0.020345	509.167600
3	0.037084	535.166034
4	0.052290	556.618323
5	0.071386	579.357098
6	0.092945	607.319837
7	0.114836	631.733899
8	0.136258	658.225063

$$\varepsilon_t = \ln(1 + \varepsilon_e)$$

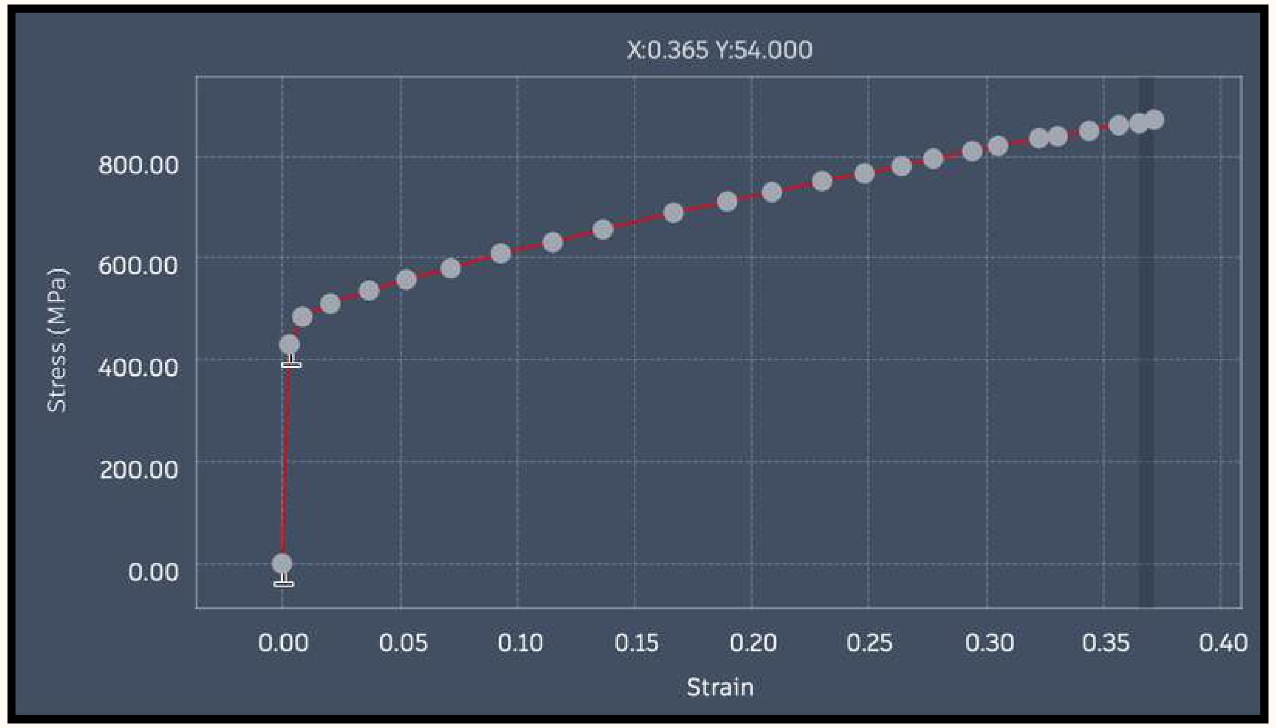
True Strain Formula

$$\sigma_t = \sigma_e(1 + \varepsilon_e)$$

True Stress Formula

Custom Material

AM Stainless Steel 316L - Nonlinear



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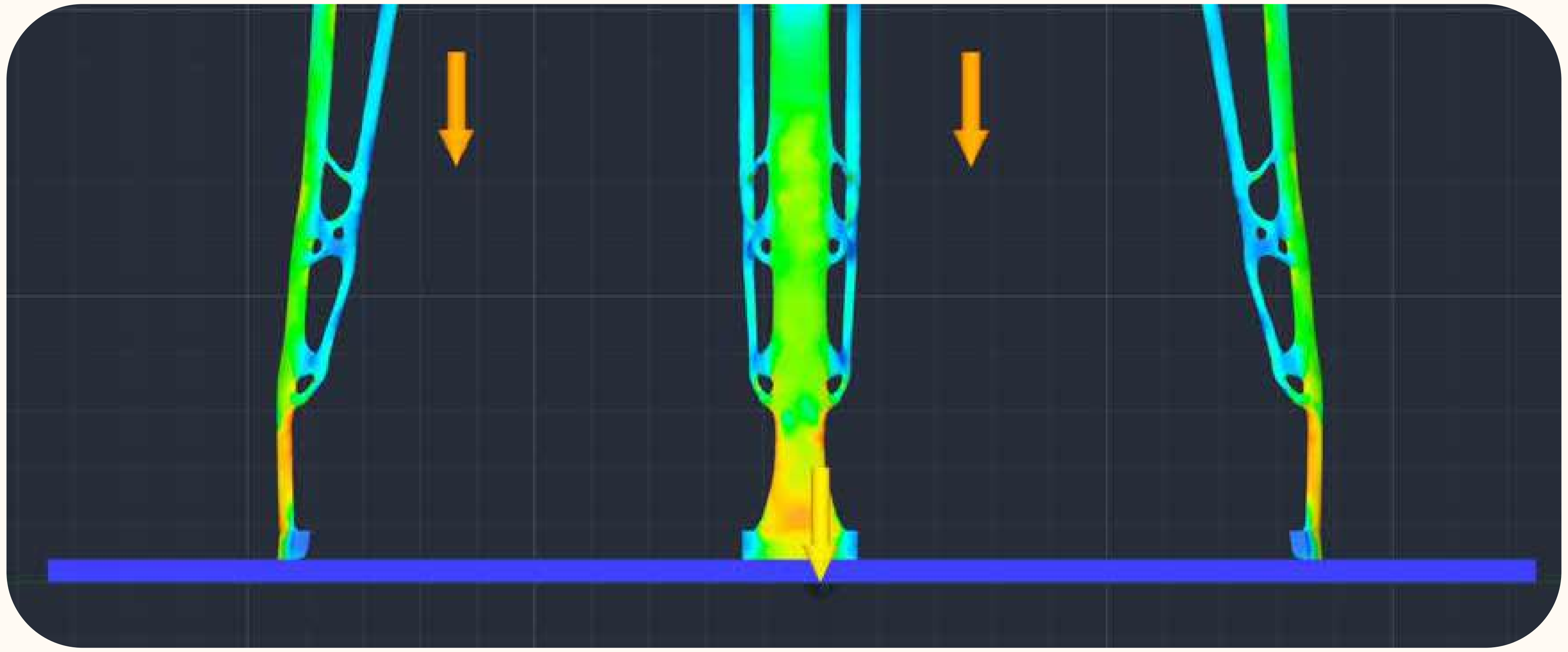
TECHNICAL INFORMATION

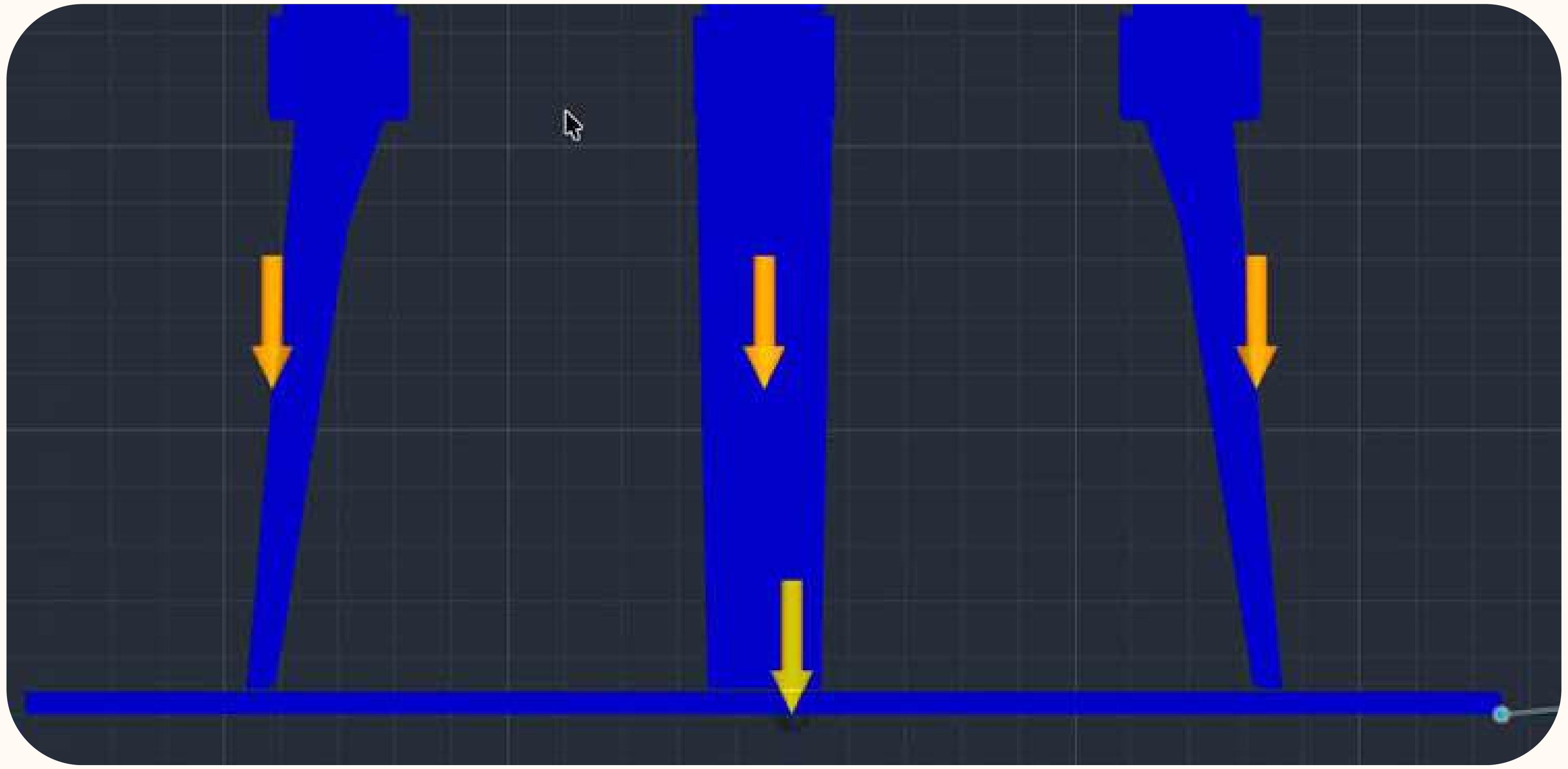
Material: 316L Stainless Steel
Drop Heights: 0.15m, 0.3m, and 0.5m
Velocities: 1.72 m/s, 2.43 m/s, and 3.13 m/s
Impact Surface: Rigid Floor
Constraints: Structural Constraint
Contacts: Separated
Gravity: Enabled

$$v = \sqrt{2gh}$$

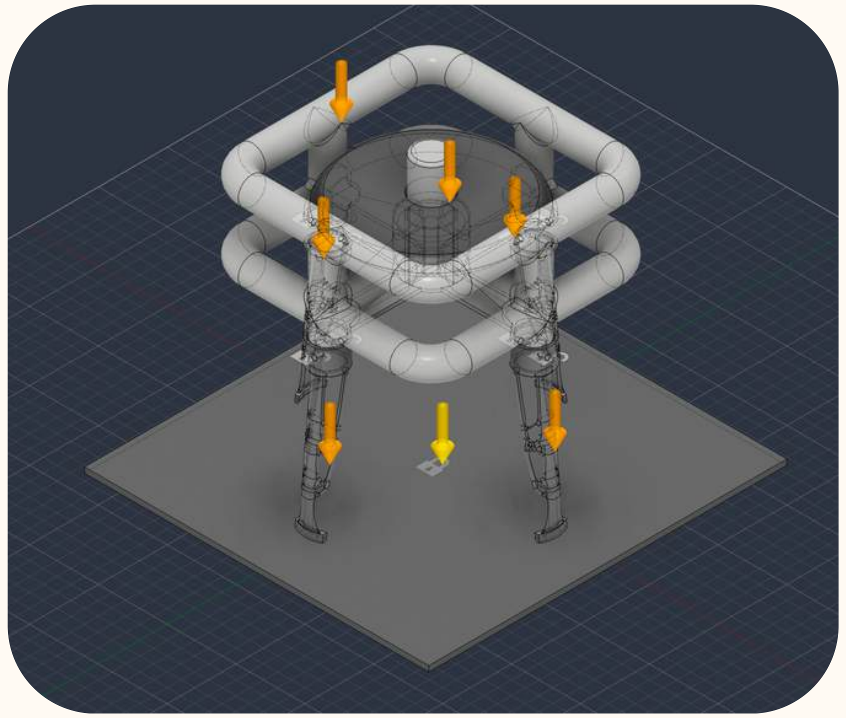
SETUP



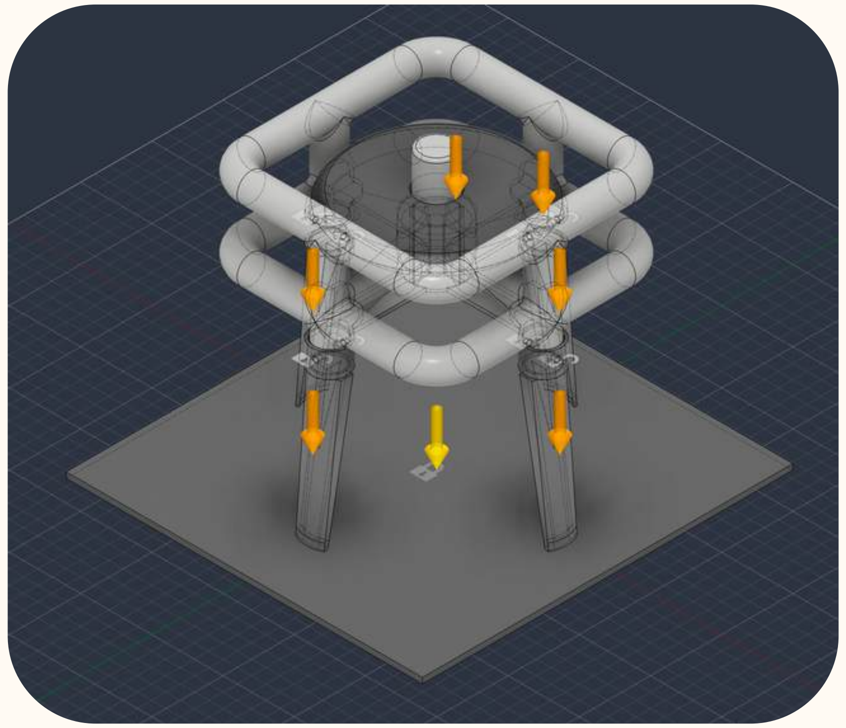
Optimized Design



Original design



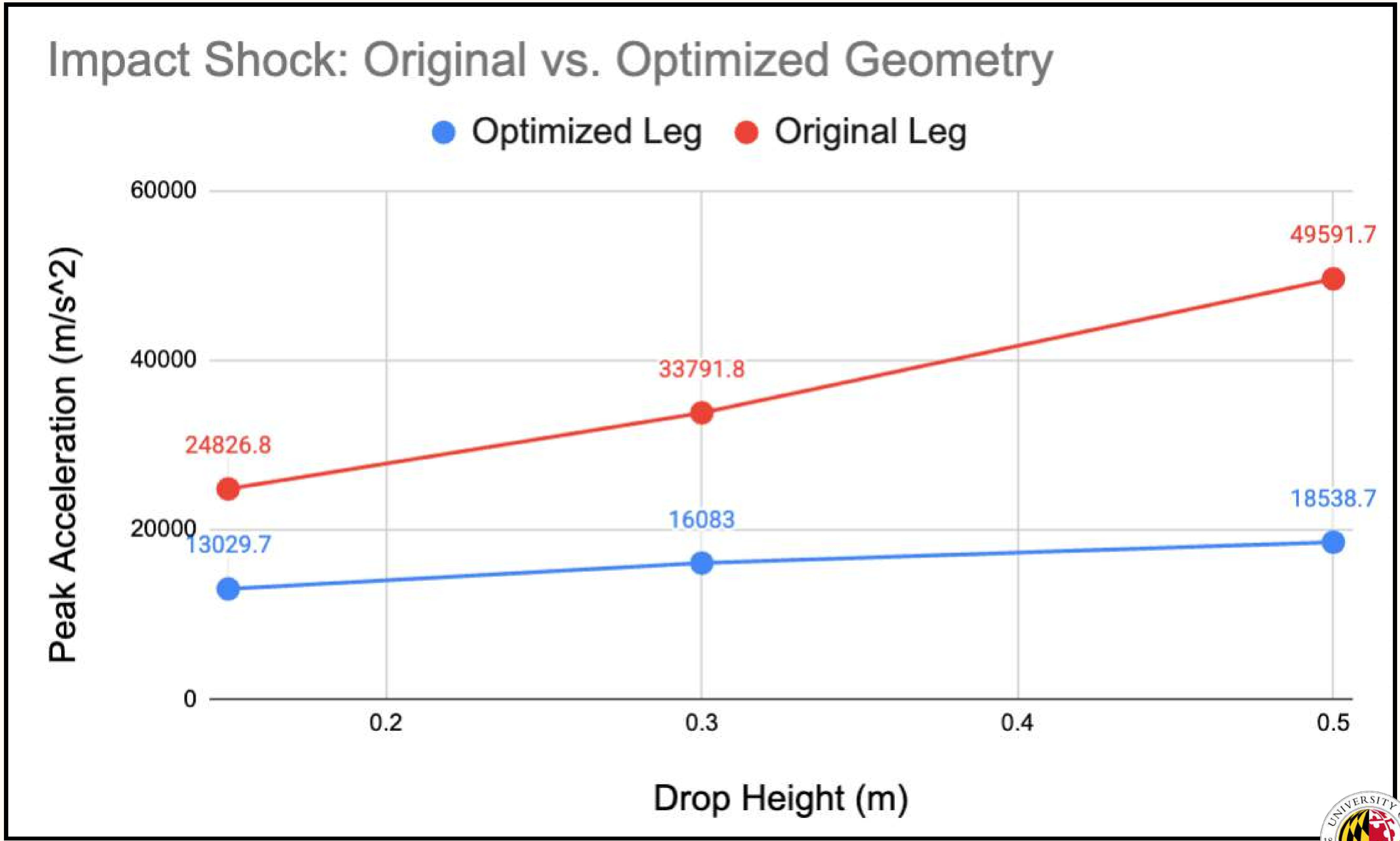
Optimized Design



Original design

FINDINGS

A 62% reduction in Peak acceleration at 0.5 m



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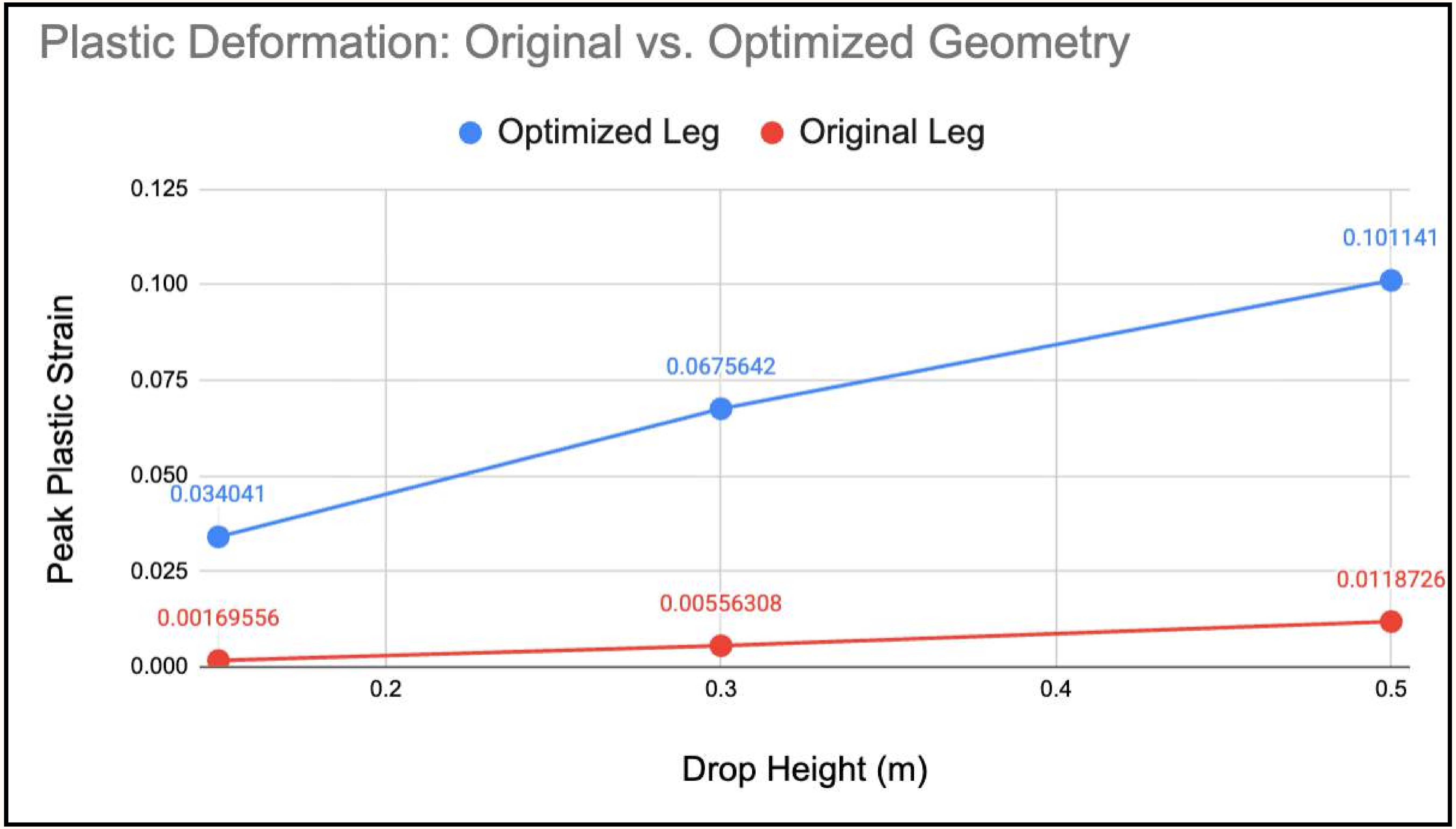
FINDINGS

10.1% Max Yielding



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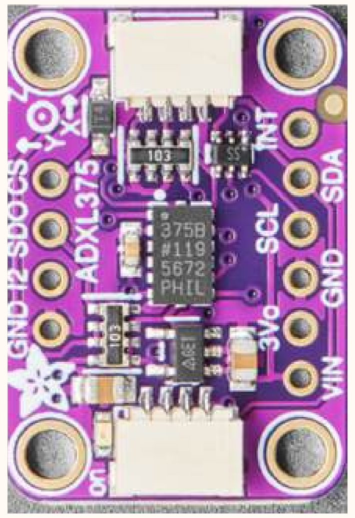
ELECTRONICS

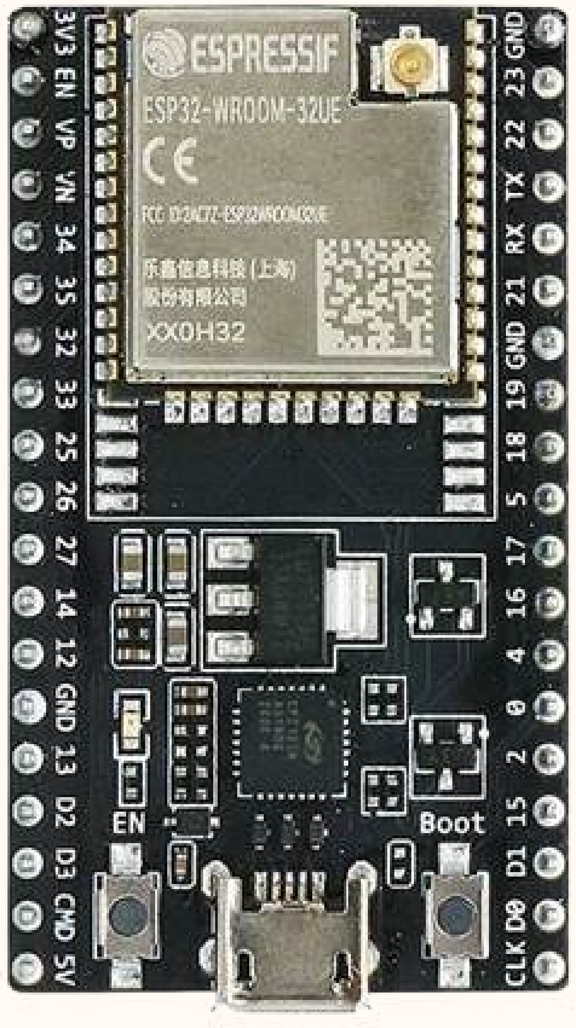
KEY METRICS

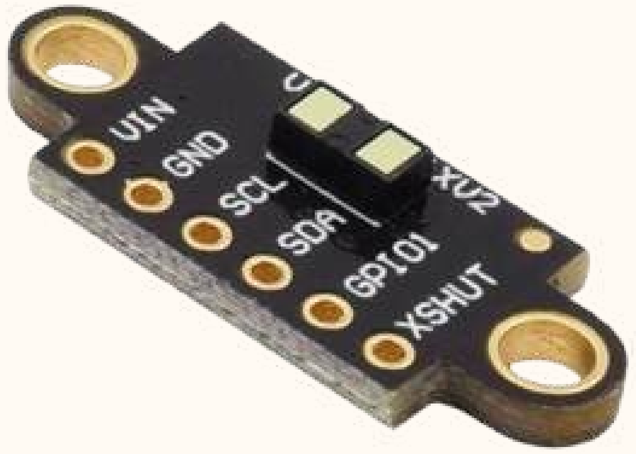
- Fall duration; <10 sec
- Free fall
- Landing Velocity
- Peak Impact force
- Bounce Height
- Structural failure Impact
- Distrance from ground during drop tests..

Selected Components

- ESP32- Microcontroller for Data Collection
- ADXL375(High G Accelerometer)
  - Measures X,Y,Z acceleration
  - Wide measurement range (0-200g) suitable for high--impact events.
  - Reliable for testing spikes.
- VL531X- Distance sensor







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ESP32 I2C COMM ⟷

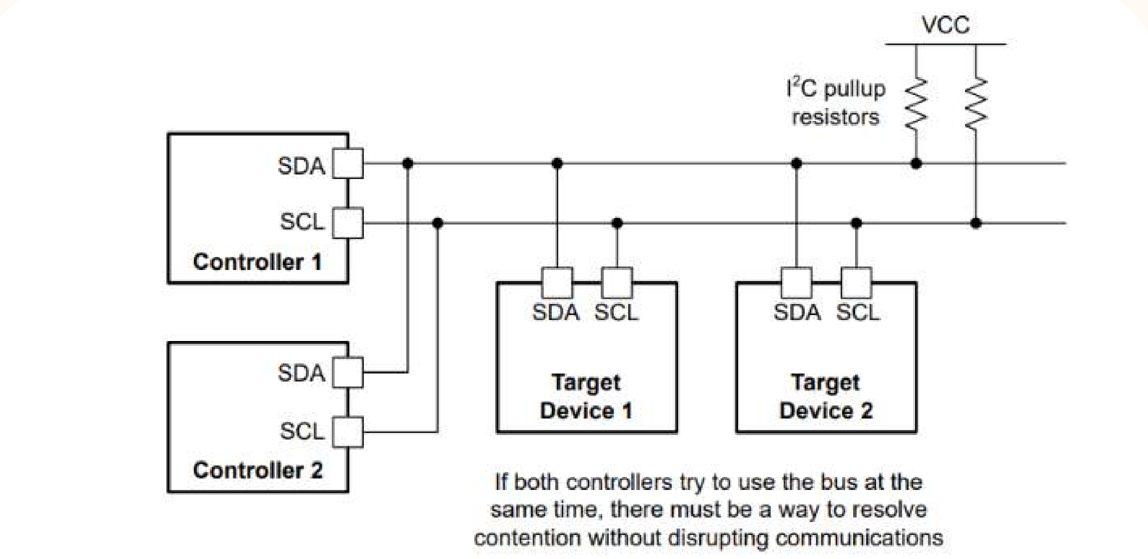


Figure 6-1. I²C Bus Contention With Multiple Controllers

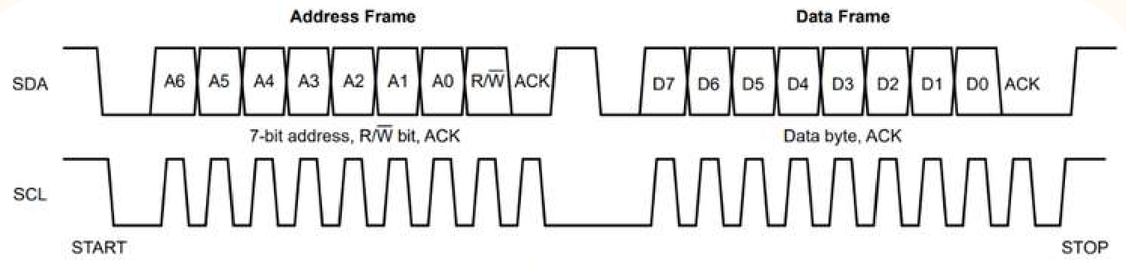


Figure 3-3. I²C Address and Data Frames

I2C (inter Integrated circuit) is a communication protocol that allows a microcontroller(master) communicate with multiple sensors (slave) using only two signal wires.

WHY we use I2C?
Built into the Arduino <Wire.h> Library
Multiple sensors share the same bread board bus.
Uses only two communication wires.

- SCL → Serial Clock synchronization
- SDA → Data Transfer between ESP32 and sensors
- VIN/VCC → 3V or 5V

Reliable for real-time sensor measurements

Communication sequence

- ESP32
- Send Sensor REGISTERS/ADDRESS- Stores 8 bit data
- Sensor acknowledges (ACK)
- Read/Write Register - //writeRegister(), //readRegister()
- Transfer Data/ Print
- STOP

"3-Axis, ±200 G Digital MEMS Accelerometer." n.d. Accessed July 30, 2026.

Sensor Calibration;

To improve mesaurement accurace, sensor offsets were calculated before testing;

Calibration Process

1. Calibrated distance sensor to zero out the baseline distance from the foot to the ground for precise measurements.

```
float rawDistanceCm = (distanceMm / 10.0);

// Remove sensor-to-foot mounting offset
float correctedDistanceCm = rawDistanceCm - SENSOR_OFFSET_CM;
```

2. Calculated multiple stationary readings, Calculated the average X,Y, Z values.
3. Determined offset values using;

Offset = Measured Values – Value Average

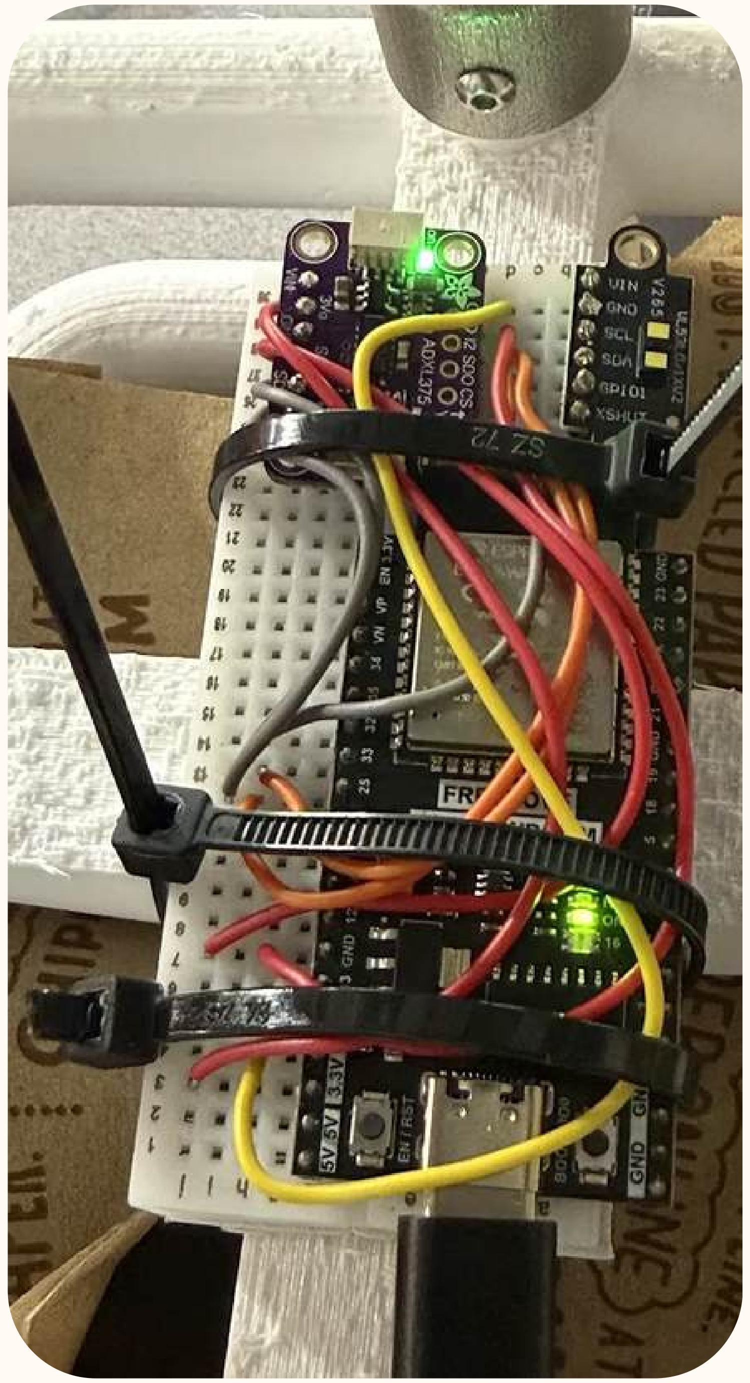
4. Applied offsets in software to correct future measurements.

```
if (readAcceleration(rawX, rawY, rawZ)) {
  accelX = (rawX * ADXL_LSB_G * GRAVITY_MS2) - 3.17;//X0 = 3.17
  accelY = (rawY * ADXL_LSB_G * GRAVITY_MS2) - 0.38; //Y0 = 0.38
  accelZ = (rawZ * ADXL_LSB_G * GRAVITY_MS2) - 2.0; //z0 = -2.15
```

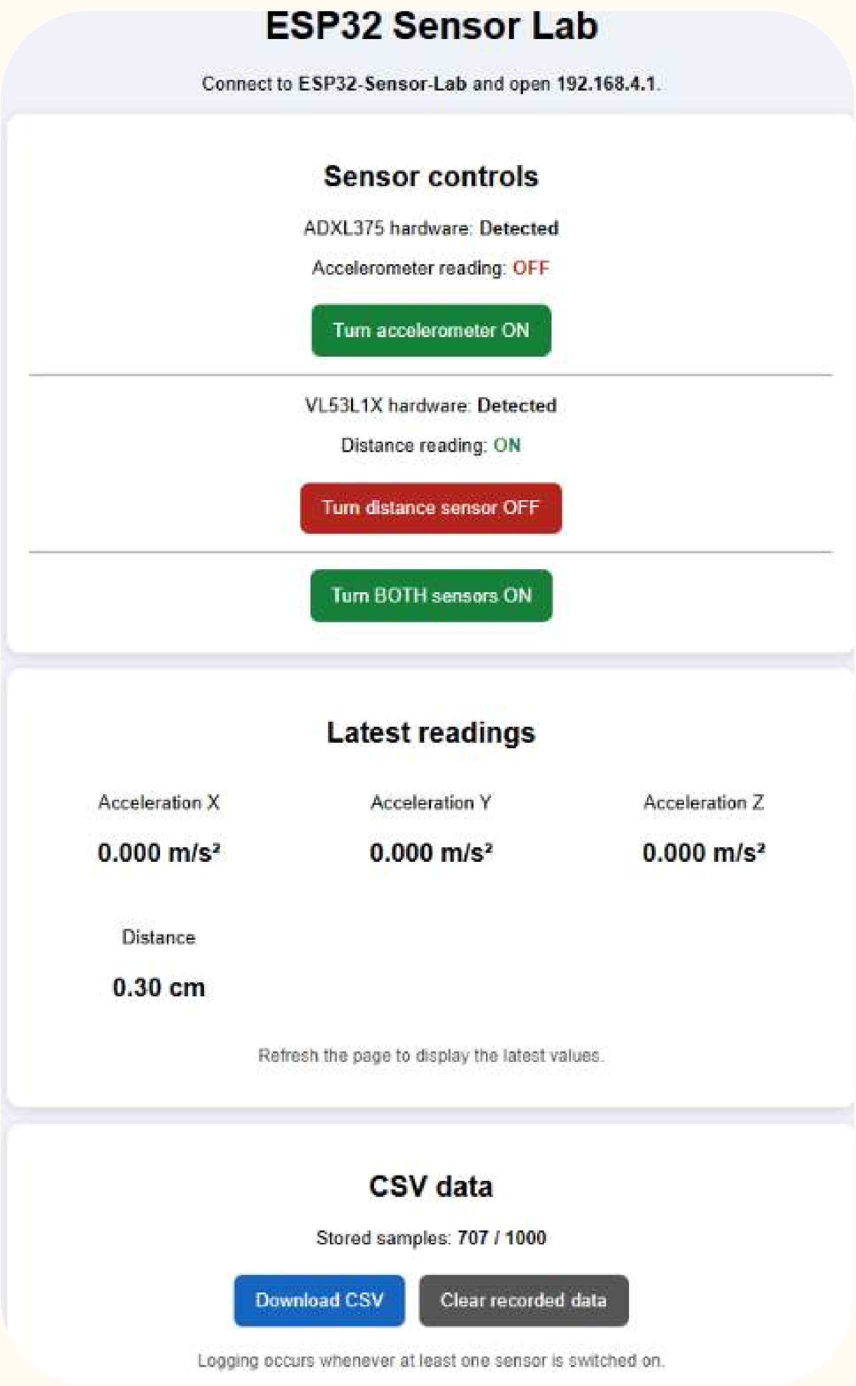
Results: Reduced errors and improved accuracy.

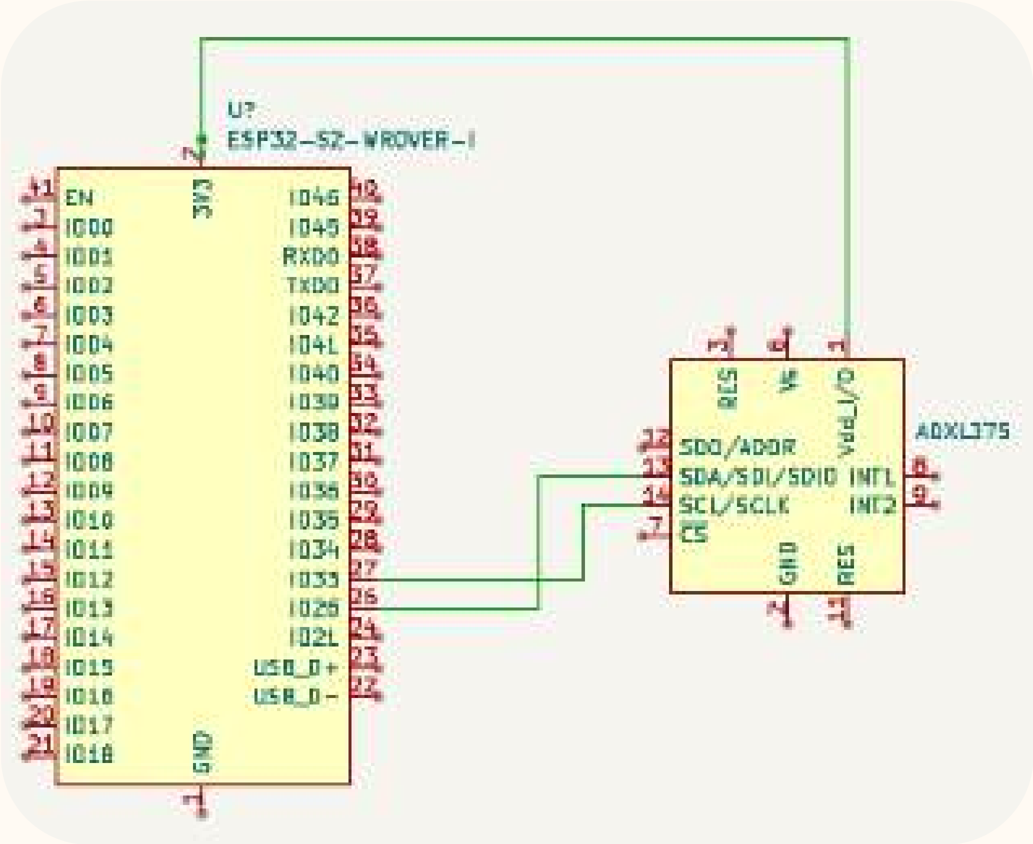
"3-Axis, ±200 G Digital MEMS Accelerometer." n.d. Accessed July 30, 2026.

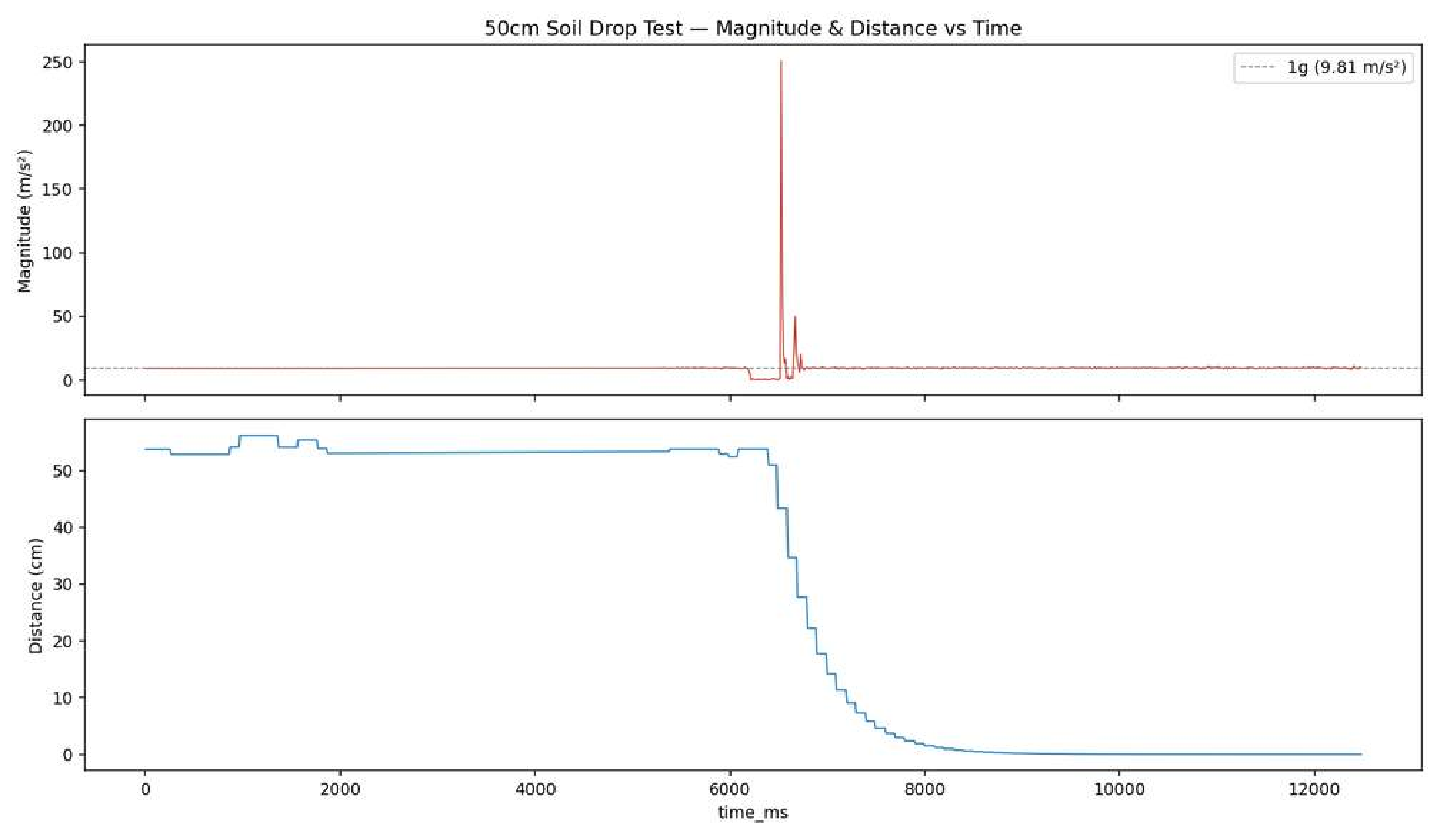
WIRING



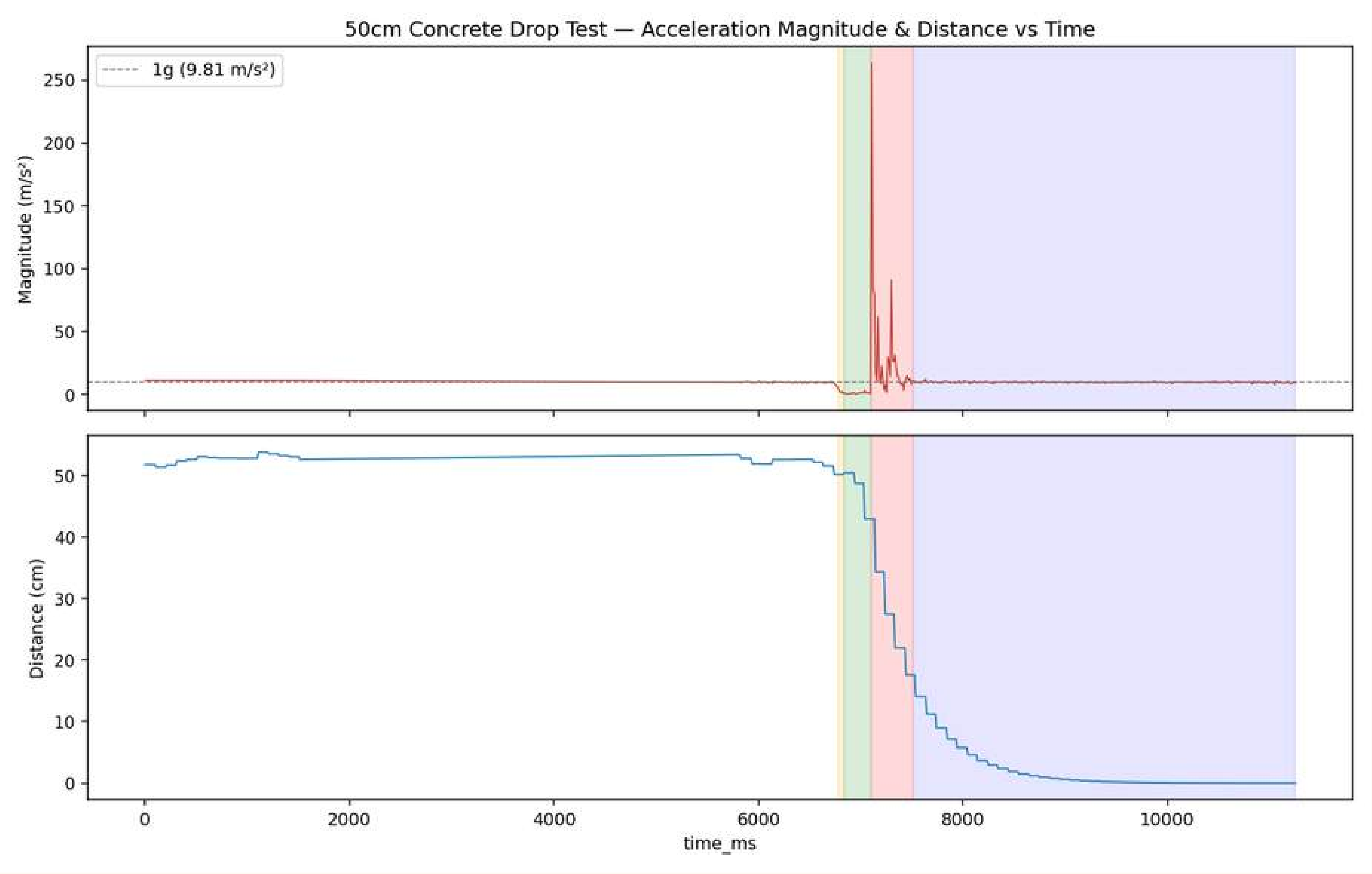
WEB SERVER



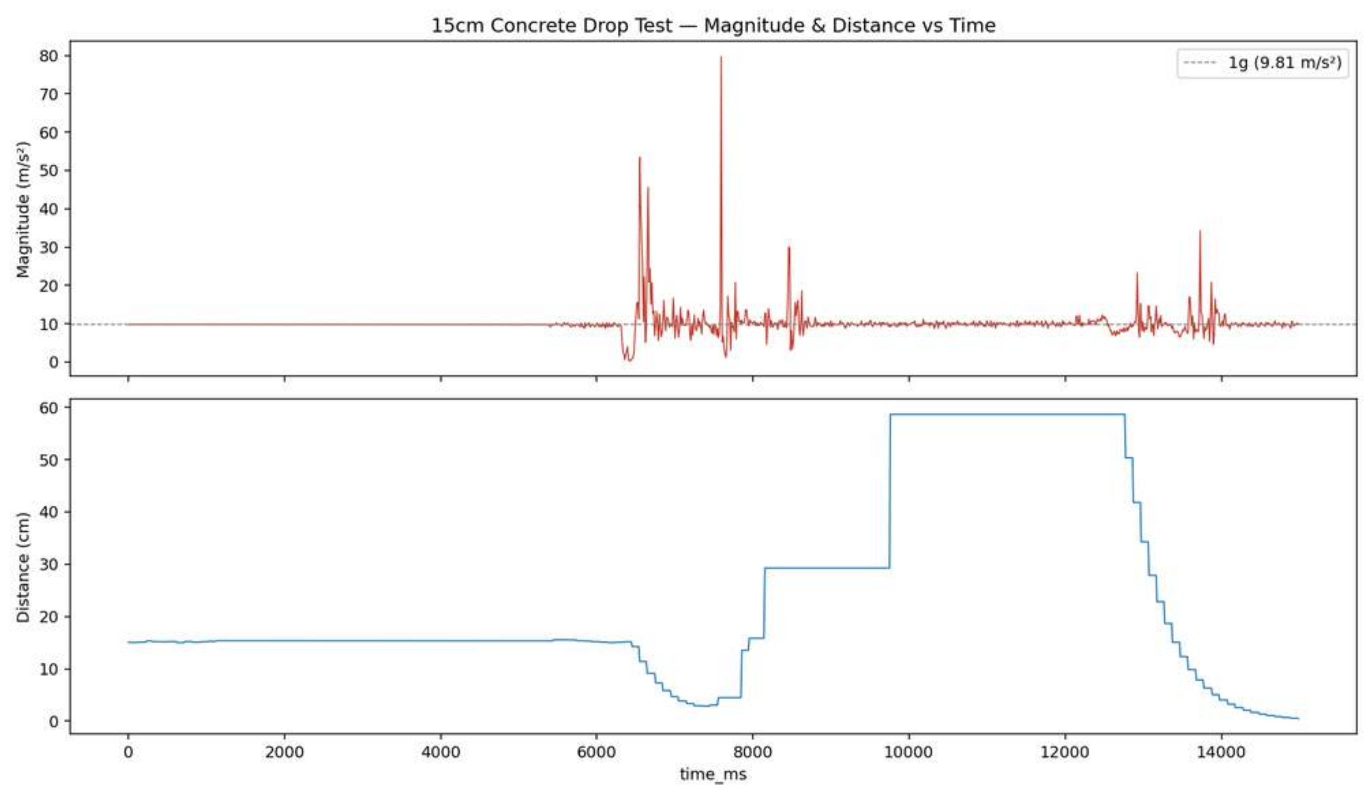




50CM SOIL DROPTEST



50CM CONCRETE DROP TEST



15CM CONCRETE DROP TEST

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FABRICATION

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RIG FABRICATION

- Material: PETG
- Method: FDM (Bambulab X1 Carbon)
- Layer Lines: .16
- First Printed Horizontally
- The rest is printed Vertically
- Infill types: Gyroid

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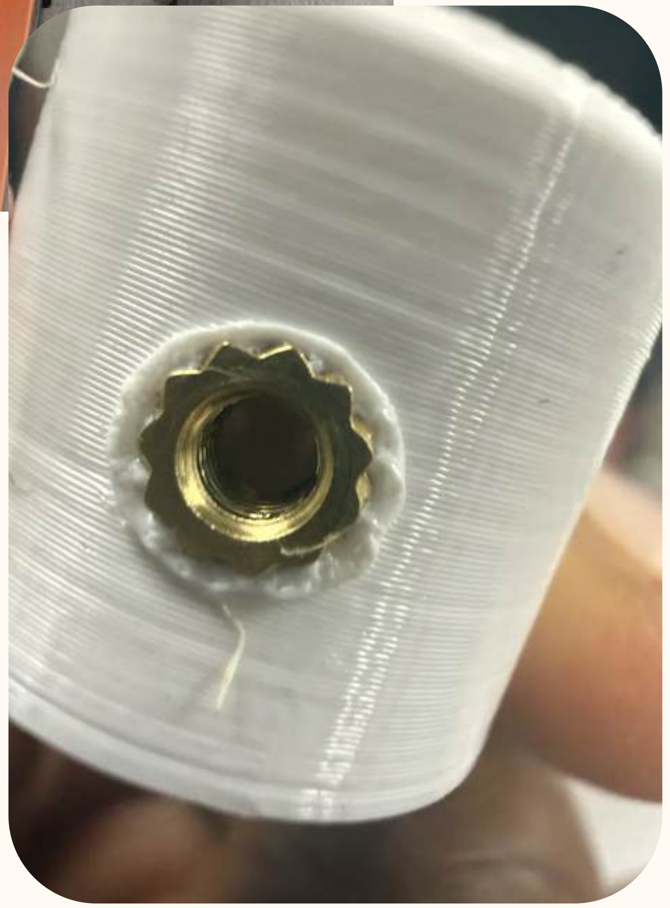
NYLON FOOT FABRICATION

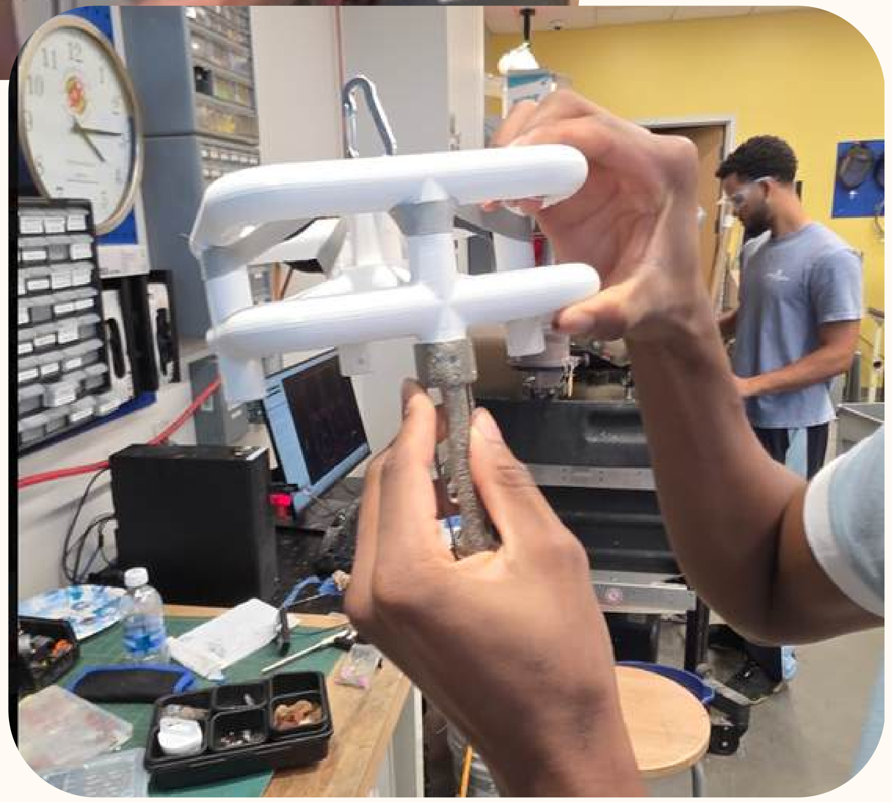
- Material: Nylon 12
- Method: SLS (Formlabs Fuse)
- Layer lines: Negligible more isotropic

STEEL FOOT FABRICATION

- Material: Steel
- Method: LPBF (EOS gmbh)
- Layer lines: Negligible more isotropic

Post Processing





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DROP TESTING

Camera Tracking

Electronics: Accelerometer and distance sensor

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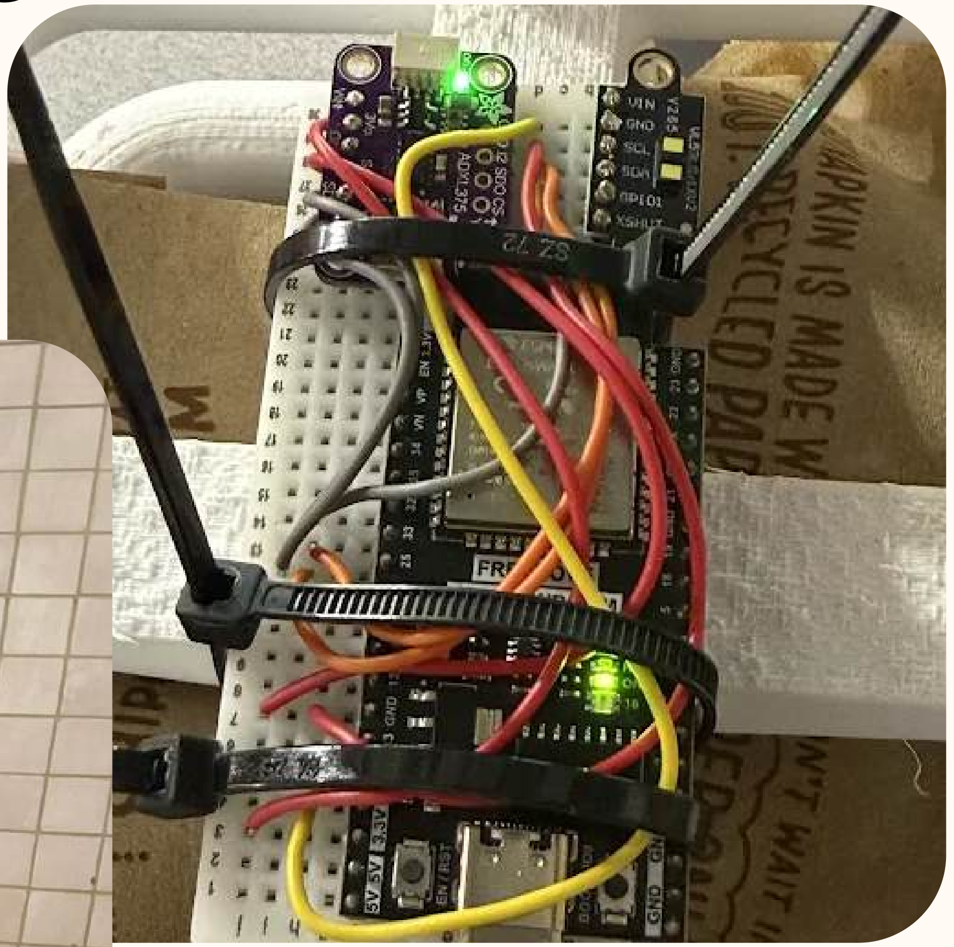
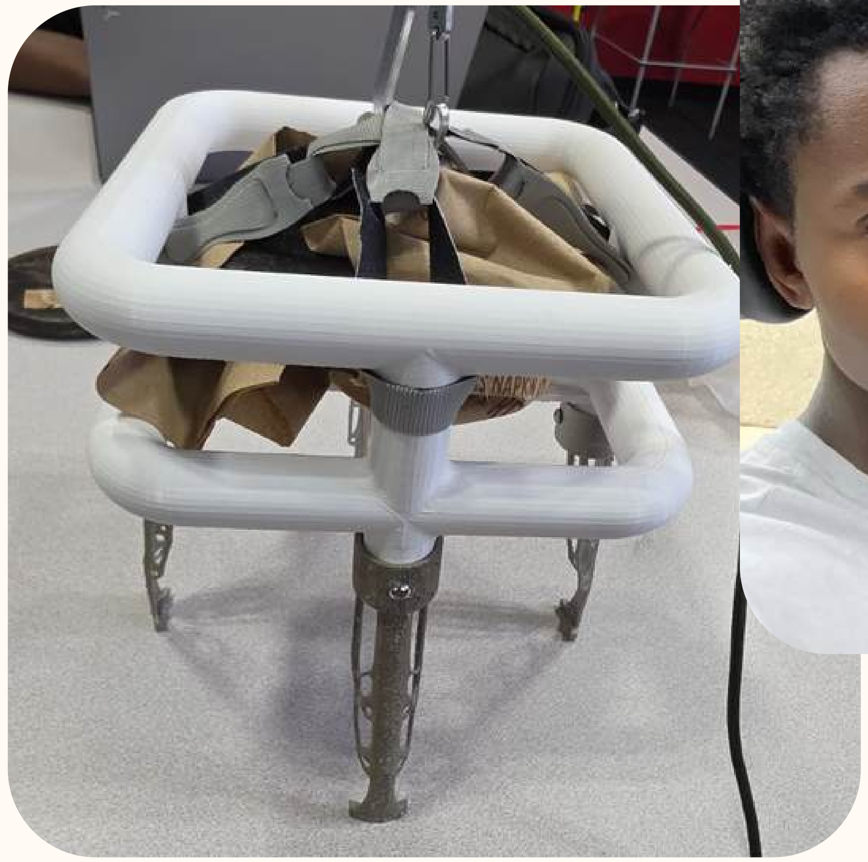
RESULTS

Preparation and Setup





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DROP TEST

See in real time how different designs perform

- Coordinate group preparation
- Release a complete model using a pulley
- Gather visual and graphical data
- Use gathered information to inform a conclusion

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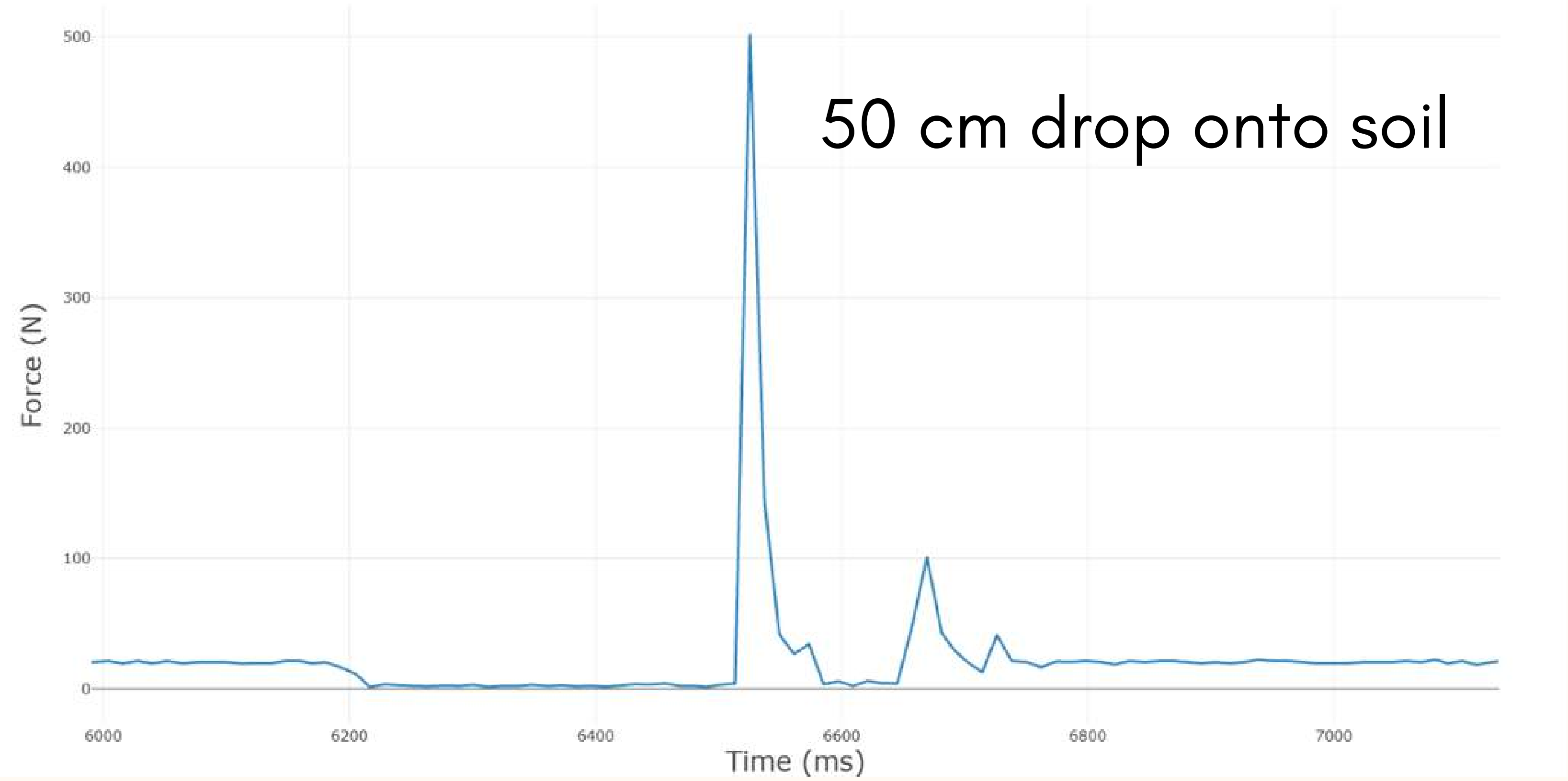
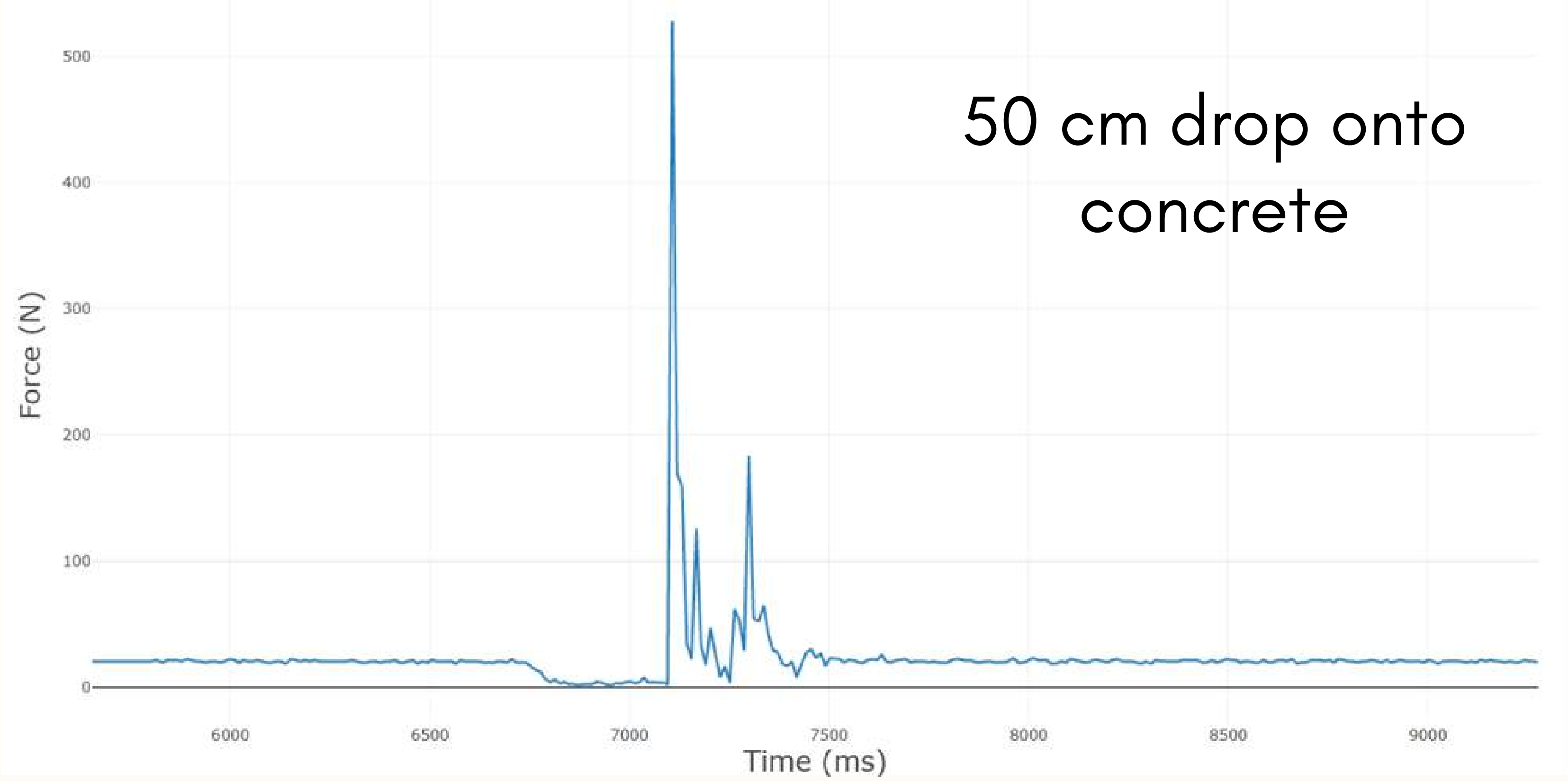
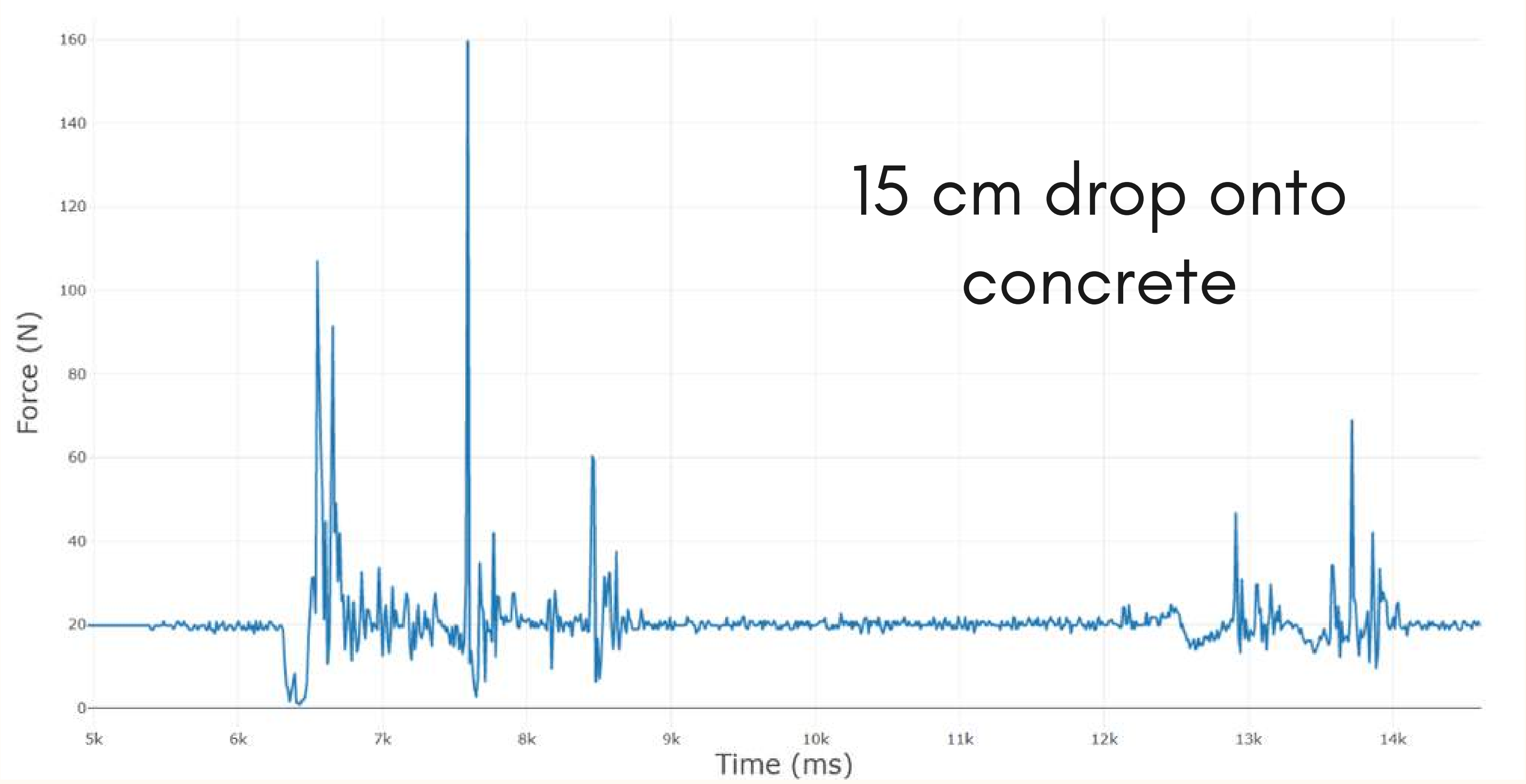
GRAPHICAL DATA

Compare felt force as a function of time







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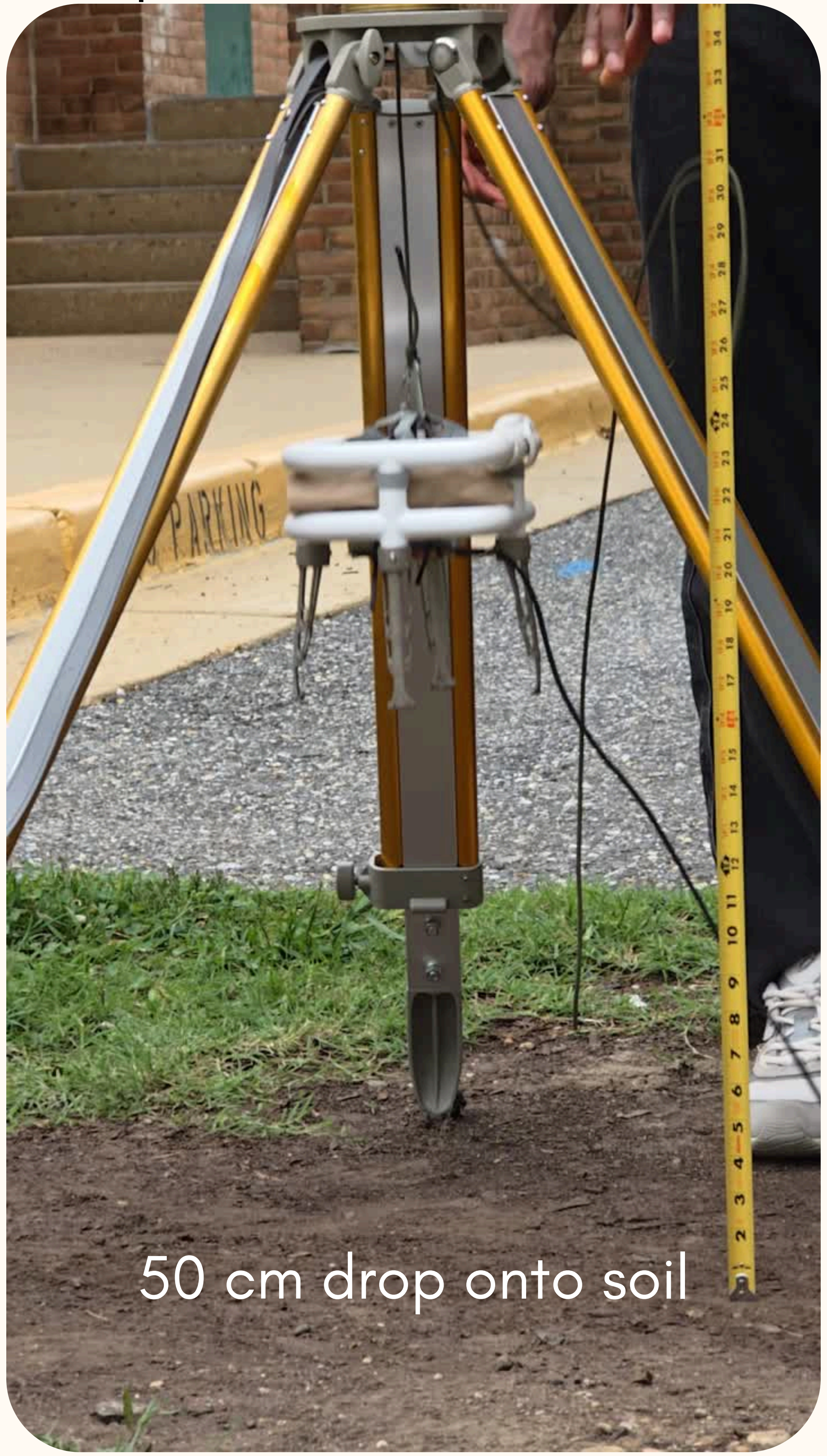
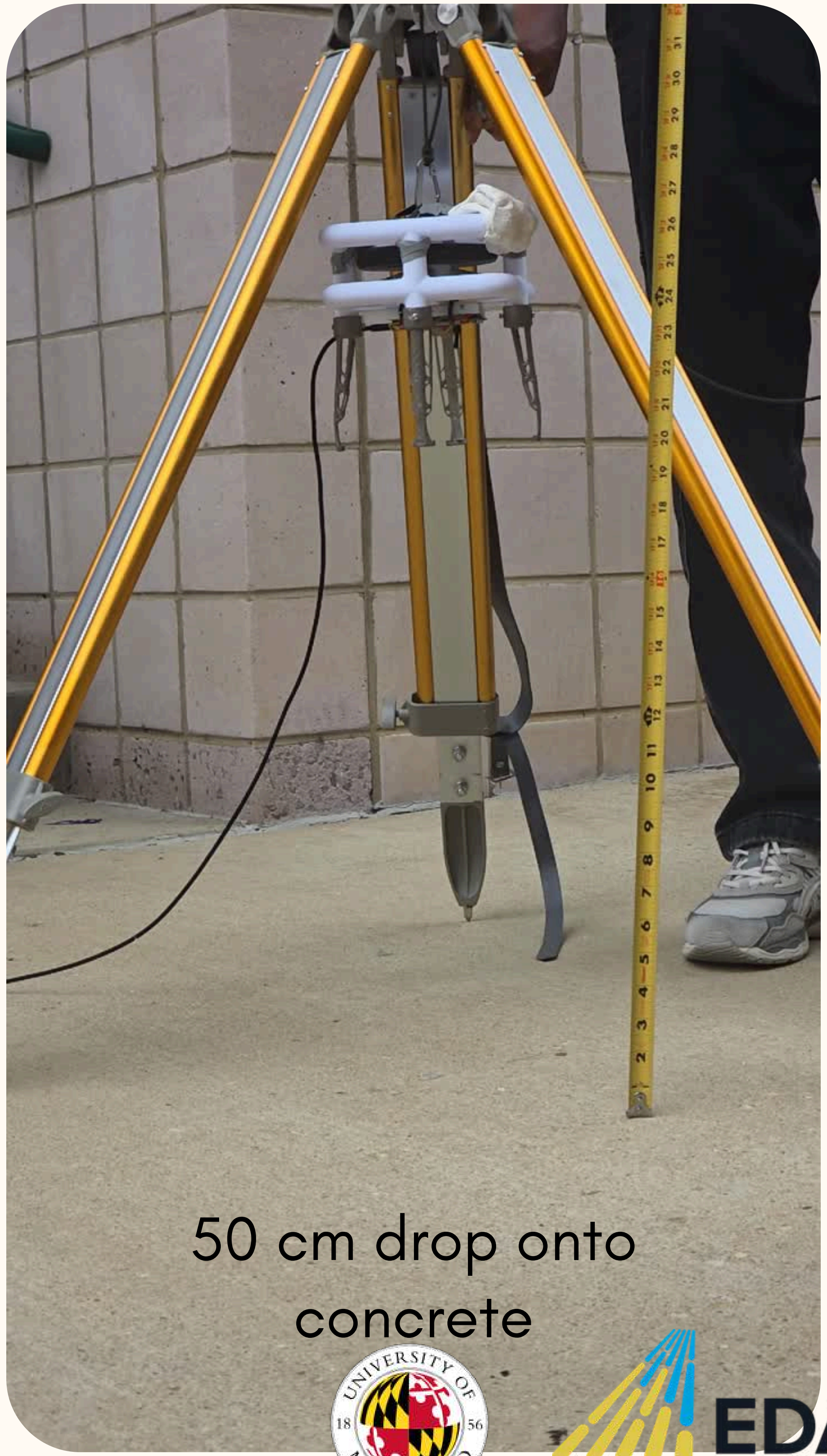
VISUAL DATA

Compare felt force as a function of time



50 cm drop onto soil

15 cm drop onto concrete



50 cm drop onto concrete

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CONCLUSION

HELPFUL LINKS

- *https://how2electronics.com/*
- *https://howtomechatronics.com/*
- *https://www.instructables.com/*
- *https://docs.arduino.cc/*
- *https://docs.cirkitdesigner.com/*
- *https://randomnerdtutorials.com/*